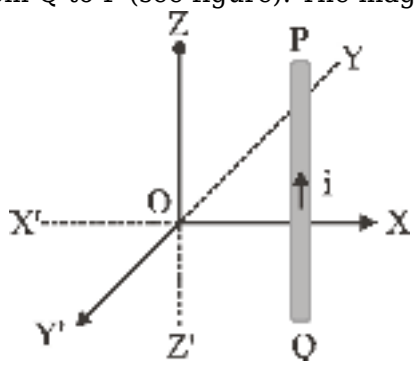
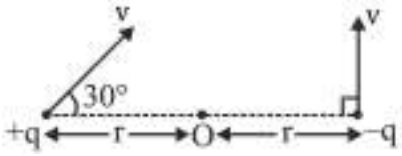
PHYSICS

1) A vertical wire kept in Z-X plane carries a current from Q to P (see figure). The magnetic field due to current will have the direction at the origin O along :



(1) OX
(2) OX'
(3) OY
(4) OY'

2) At any instant, two charge particle +q and –q are moving as shown in figure. Find the magnitude and direction of magnetic field at point 'O'.



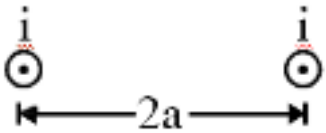
(1) $\frac{\mu_0 qv}{8\pi r^2}\otimes$
(2) $\frac{\mu_0 qv}{4\pi r^2}\odot$
(3) $\frac{3\mu_0 qv}{4\pi r^2}\otimes$
(4) $\frac{3\mu_0 qv}{8\pi r^2}\otimes$

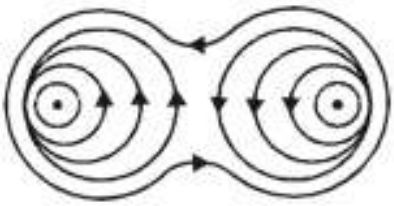
3) An electron moves in a circular orbit with a uniform speed v. It produces a magnetic field B at the centre of the circle. The radius of the circle is proportional to

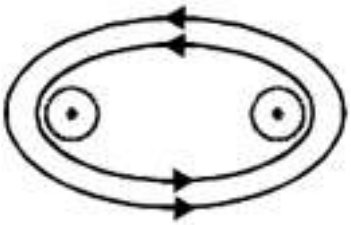
(1) B/v
(2) v/B
(3) $\sqrt{\frac{v}{B}}$
(4) $\sqrt{\frac{B}{v}}$

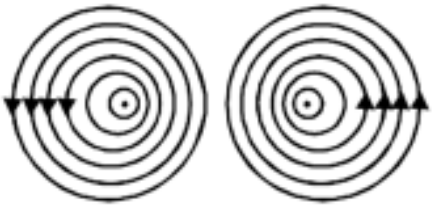
4) Which of the following field patterns is correct for two long straight equal parallel current

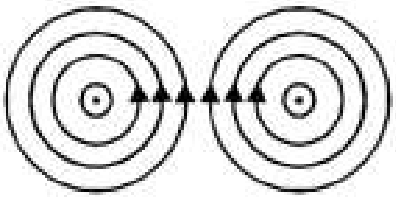
carrying wires ?



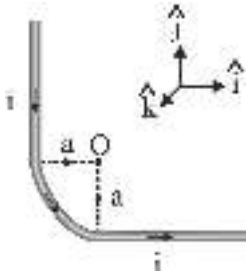
(1) 

(2) 

(3) 

(4) 

5) The unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$ are as shown below. What will be the magnetic field at O in the following figure :-



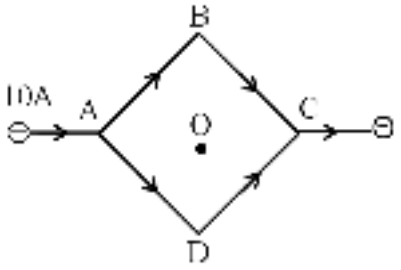
(1) $\frac{\mu_0}{4\pi}\frac{i}{a}\left(2-\frac{\pi}{2}\right)\hat{j}$

(2) $\frac{\mu_0}{4\pi}\frac{i}{a}\left(2+\frac{\pi}{2}\right)\hat{j}$

(3) $\frac{\mu_0}{4\pi}\frac{i}{a}\left(2+\frac{\pi}{2}\right)\hat{i}$

(4) $\frac{\mu_0}{4\pi}\frac{i}{a}\left(2+\frac{\pi}{2}\right)\hat{k}$

6) Figure shows a square loop with edge length 'a' meter. The resistance of wire ABC is 'r' Ω and



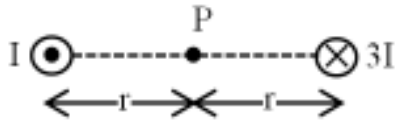
that of ADC is '2r' Ω. Magnetic field at the centre of the loop is :-

(1) $\frac{10\sqrt{2}\mu_0}{3\pi a}\otimes$

(2) $\frac{10\sqrt{2}\mu_0}{3\pi a}\odot$

(3) $\frac{\sqrt{2}\mu_0}{6\pi a}\otimes$

(4) $\frac{\sqrt{2}\mu_0}{6\pi a}\odot$



7) Find magnetic field at point P.

(1) $\frac{\mu_0 I}{\pi r}$

(2) $\frac{\sqrt{2}\mu_0 I}{\pi r}$

(3) $\frac{\sqrt{3}\mu_0 I}{\pi r}$

(4) $\frac{2\mu_0 I}{\pi r}$

8)

A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields **B** and **B'** at radial distances $\frac{a}{2}$ and 2a respectively, from the axis of the wire is :-

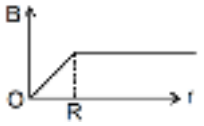
(1) $\frac{1}{2}$

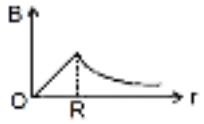
(2) 1

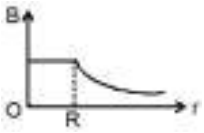
(3) 4

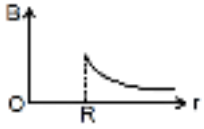
(4) $\frac{1}{4}$

9) An infinitely long hollow conducting cylinder with radius R carries a uniform current along its surface. Choose the correct representation of magnetic field (B) as a function of radial distance (r) from the axis of cylinder.

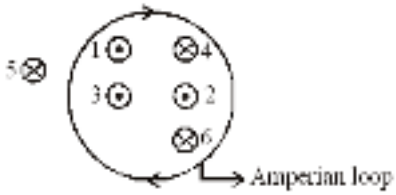
(1) 

(2) 

(3) 

(4) 

10) The figure shown has a hypothetical amperian loop with the direction of movement marked. Six infinitely long wires having direction of current as shown and magnitude as follows are placed perpendicular to the loop :-



I_1 = 1A I_4 = 1A
I_2 = 2A I_5 = 5A
I_3 = 3A I_6 = 4A

The value of $\oint \vec{B}.\vec{dl}$ around the closed amperian loop will be :-

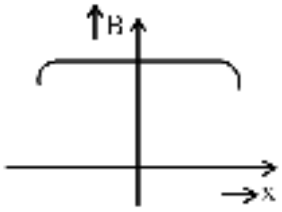
(1) μ_0 Wb/m

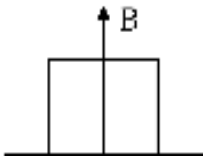
(2) $-\mu_0$ Wb/m

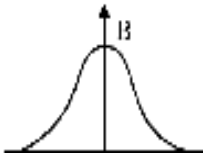
(3) $4\mu_0$ Wb/m

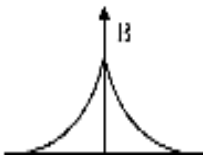
(4) $-4\mu_0$ Wb/m

11) B along the axis of a solenoid is given by which of the following diagrams ?

(1) 

(2) 

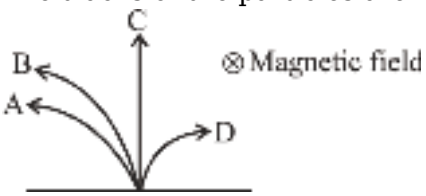
(3) 

(4) 

12) A wire is wound on a long rod of material of relative permeability $\mu_r = 4000$ to make a solenoid. If the current through the wire is 5 A and number of turns per unit length is 1000 per metre, then the magnetic field inside the solenoid is :

(1) 25.12 mT
(2) 12.56 mT
(3) 12.56 T
(4) 25.12 T

13) A neutron, a proton, an electron and an α-particle enter a region of uniform magnetic field with the same velocities. The magnetic field is perpendicular and directed into the plane of the paper. The tracks of the particles are labelled in the figure. The electron follows the track .....



(1) A
(2) B
(3) C
(4) D

14) Work done by magnetic force on moving charge is

(1) Zero
(2) Positive
(3) Negative
(4) May be ≥ 0

15) When a charged particle enters in a uniform magnetic field its kinetic energy :-

(1) remains constant
(2) increases
(3) decreases
(4) becomes zero

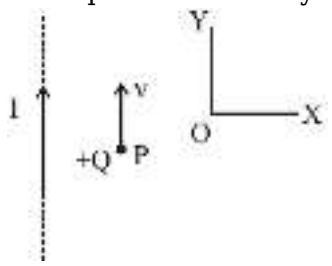
16) An electron (mass = 9.0×10^{-31} kg and charge = 1.6×10^{-19} coulomb) is moving in a circular orbit in a magnetic field of 1.0×10^{-4} Wb/m^2. Its period of revolution is ..... .

(1) 3.5×10^{-7} s
(2) 7.0×10^{-7} s

(3) 1.05×10^{-6} s

(4) 2.1×10^{-6} s

17) A charge Q moves parallel to a very long straight wire carrying a current I as shown. The force



on the charge is

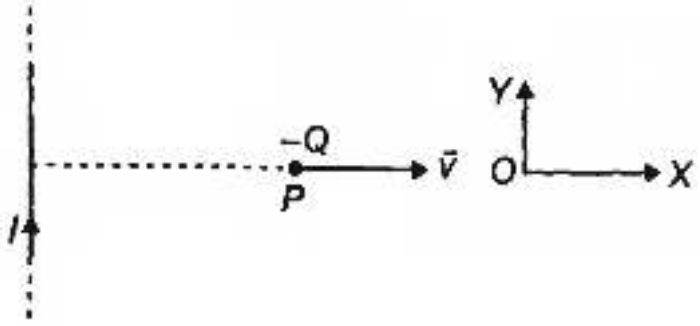
(1) Opposite to OX

(2) Along OX

(3) Opposite to OY

(4) Along OY

18) A very long straight wire carries a current I. At the instant when a charge -Q at point P has



velocity $\vec{v}$-, as shown the force on the charge is :

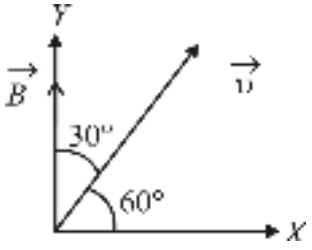
(1) Opposite to OX

(2) Along OX

(3) Opposite to OY

(4) Along OY

19) A proton of mass 1.67×10^{-27}kg and charge 1.6×10^{-19}C is projected with a speed of 2×10^{6} m/s at an angle of 60° to the X-axis as shown. If a uniform magnetic field of 0.104 Tesla is applied along Y-axis, the path of proton is



(1) A circle of radius 0.2 m and time period $\pi \times 10^{-7}$s

(2) A circle of radius 0.1 m and time period $2\pi \times 10^{-7}$s

(3) A helix of radius 0.1 m and time period $2\pi \times 10^{-7}$s

(4) A helix of radius 0.2 m and time period $4\pi \times 10^{-7}$s

20) A charged particle enters a uniform magnetic field with velocity vector at an angle of 45° with

the magnetic field. The pitch of the helical path followed by the particle is p. The radius of the helix will be :-

(1) $\frac{p}{\sqrt{2}\pi}$

(2) $\sqrt{2}p$

(3) $\frac{p}{2\pi}$

(4) $\frac{\sqrt{2}p}{\pi}$

21) H^+, He^+ and O^{++} all having the same kinetic energy pass through a region with is a uniform magnetic field perpendicular to their velocity. The masses of H^+, He^+ and O^{++} are 1 amu, 4 amu and 16 amu respectively then :-
(a) H^+ will be deflected most
(b) O^{++} will be deflected most
(c) He^+ and O^{++} will be deflected equally
(d) all will be deflected equally

(1) a, b

(2) a, b, c

(3) only a

(4) a, c

22) The radius of helical path followed by the charge (m, q_0) moving in magnetic field B; with some velocity υ is (θ is angle between $\vec{v}$ and $\vec{B}$):

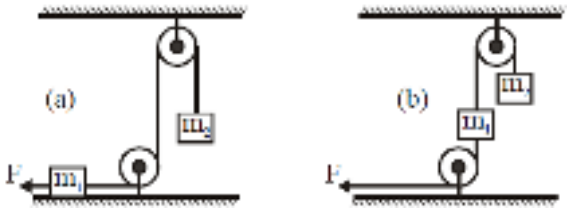
(1) $\frac{m\upsilon\sin\theta}{q_0B}$

(2) $\frac{m\upsilon\cos\theta}{q_0B}$

(3) $\frac{m\upsilon}{q_0B}$

(4) $\frac{m\upsilon\tan\theta}{q_0B}$

23) The ratio of tensions in the string connected to the block of mass m_2 in figure-(a) and figure-(b) respectively is (friction is absent everywhere) :
[m_1 = 2 kg, m_2 = 8 kg and F = 50 N].



(1) 7 : 9

(2) 9 : 7

(3) 2 : 7

(4) 7 : 2

24) Ratio of weight of a man in a stationary lift and weight of a man in a lift moving downward with an acceleration is 3 : 2 then acceleration of lift is :

(1) g / 3
(2) g / 2
(3) g
(4) 2g

25) A force is applied to two different objects. It produces accelerations of 3 m/s^2 and 6 m/s^2 respectively in these objects. If the same force is applied to the combination of these objects then acceleration of combination will be :-

(1) 2 m/s^2
(2) 4.5 m/s^2
(3) 4 m/s^2
(4) none

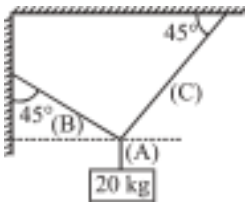
26) A ball of mass m, moving with velocity $\left(2\hat{i}+3\hat{j}\right)$ m/s recieves an impulse $\left(-3m\hat{i}\right)$ N-s due to an impact. What is the velocity of the ball just after the impact ?

(1) $\left(-\hat{i}+3\hat{j}\right)$ m/s
(2) $\left(\hat{i}-3\hat{j}\right)$ m/s
(3) $\left(-\hat{i}-3\hat{j}\right)$ m/s
(4) $\left(\hat{i}+3\hat{j}\right)$ m/s

27) Nine five rupee coins are put on the top of each other on a table. Each coin has a mass m kg. The reaction of the 5th coin (counted from bottom) on the 6th coin is :-

(1) 0 mg
(2) 2 mg
(3) 4 mg
(4) 5 mg

28) In the given diagram, the tension in string C is:-

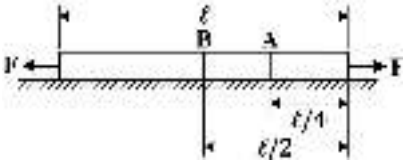


(1) 100 N
(2) 70.7 N

(3) 141 N

(4) 200 N

29) A heavy rope of mass m and length is placed on a smooth table. It is pulled at both ends by applying force F of equal magnitude as shown. Tension in the rope at point



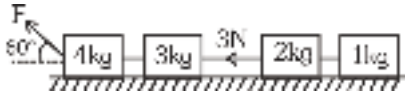
(1) A is 3F/4

(2) B is F/2

(3) A is F

(4) B is 2F

30) Figure shows four blocks that are being pulled along a smooth horizontal surface. The masses of the blocks and tension in one cord are given. The pulling force F is :-



(1) 5 N

(2) 10 N

(3) 12.5 N

(4) 20 N

31) A beaker containing some water is lying on the pan of a spring balance. If one dips finger in the water without touching its bottom, then the reading of the balance will :-

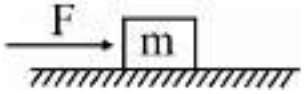
(1) increase

(2) decrease

(3) remain constant

(4) depend upon the material of beaker

32) A force F acts on a block placed on a surface with coefficient of friction μ. The magnitude of F is exactly half of that needed to just move the block. The friction force acting on the block is :



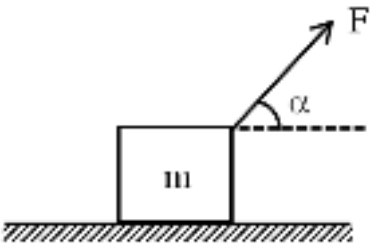
(1) 0

(2) F/2

(3) F

(4) μmg

33) At the instant t = 0 a force F = Kt (K is a constant) acts on a small block of mass m resting on a



smooth horizontal surface. The time when body leaves the surface is :

(1) $mgK\sin\alpha$

(2) $\frac{K\sin\alpha}{mg}$

(3) $\frac{mg\sin\alpha}{K}$

(4) $\frac{mg}{K\sin\alpha}$

34) At a certain instant of time the mass of a rocket going up vertically is 100 kg. If it is ejecting 5 kg of gas per second at a speed of 400 m/s, the acceleration of the rocket would be : (taking g = 10 m/s^2) :

(1) 20 m/s^2

(2) 10 m/s^2

(3) 2 m/s^2

(4) 1 m/s^2

35) A machine gun fires a bullet of mass 40 g with a velocity 1200 ms^{-1}. The man holding it can exert a maximum force of 144 N on the gun. How many bullets can he fire second at the most ?

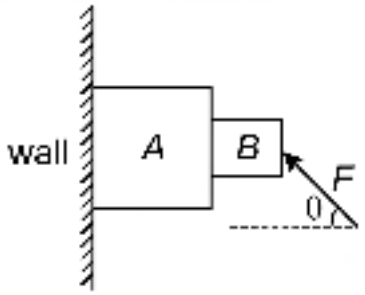
(1) One

(2) Four

(3) Two

(4) Three

36) Consider the situation shown in the figure the wall is smooth but the surfaces of blocks A and B



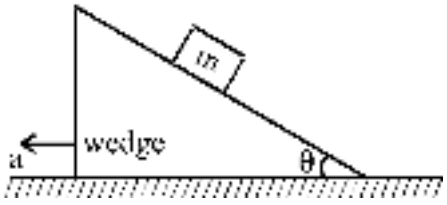
in contact are rough. the friction on B due to A in equilibrium:

(1) Is Upwards

(2) Is downwards

(3) is zero

(4) The system cannot remains in equilibrium for any value of F

37) Find the horizontal acceleration of the inclined wedge so that the block placed on the wedge falls freely :- 

(1) g tanθ

(2) g/tan θ

(3) g cos θ

(4) g sec θ

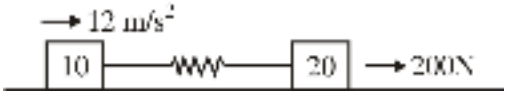
38) A block is released from the top of an inclined plane of inclination θ. Its velocity at the bottom of plane is V. If it slides down a rough inclined plane of the same inclination its velocity at bottom is V/2. The coefficient of friction is :-

(1) $\frac{3}{4}\cot\theta$

(2) $\frac{1}{4}\cot\theta$

(3) $\frac{1}{4}\tan\theta$

(4) $\frac{3}{4}\tan\theta$

39) Two masses of 10 kg and 20 kg, respectively are connected by a light spring as shown in the figure. A force of 200 N acts upon the 20 kg mass. At the instant shown the 10 kg mass has acceleration of 12 m/s^2. The acceleration of 20 kg mass is :- 

(1) 12 m/s^2

(2) 4 m/s^2

(3) 10 m/s^2

(4) Zero

40) A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m by applying a force P at the end of the rope. The force which the rope exerts on the block is :- 

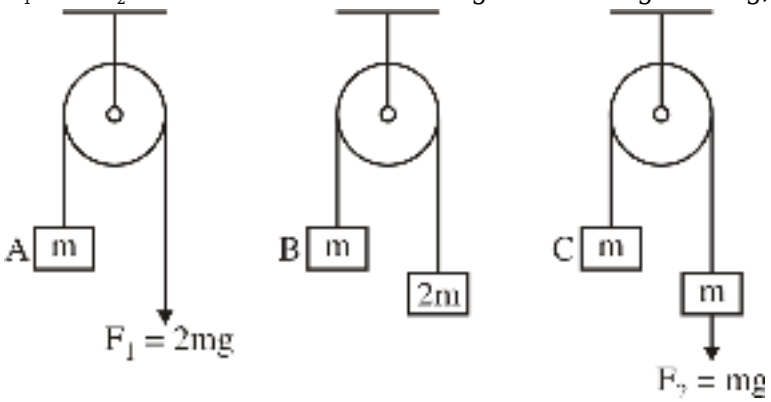
(1) $\frac{P}{M-m}$

(2) $\frac{PM}{M-m}$

(3) $\frac{Pm}{M+m}$

(4) $\frac{PM}{M+m}$

41) In the figure the blocks A, B and C of mass m each have acceleration a_1, a_2 and a_3 respectively. F_1 and F_2 are external forces of magnitudes 2mg and mg, respectively.



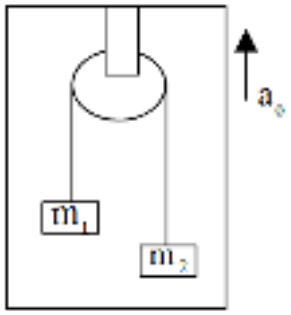
(1) $a_1 = a_2 = a_3$

(2) $a_1 > a_3 > a_2$

(3) $a_1 = a_2$, $a_2 > a_3$

(4) $a_1 > a_2$, $a_2 = a_3$

42) Consider an arrangement of pulley and blocks kept in an elevator which is moving upward with acceleration a_0. The acceleration of block m_1 w.r.t. elevator is (Assume the pulley to be massless and $m_2 > m_1$)



(1) $\frac{m_2 - m_1}{m_1 + m_2}(g - a_0)$

(2) $\frac{m_2 + m_1}{m_2 - m_1}(g - a_0)$

(3) $\frac{m_2 - m_1}{m_1 + m_2}(g + a_0)$

(4) $\frac{m_2 + m_1}{m_1 - m_2}(g + a_0)$

43) A uniform rope of length ℓ lies on a table. If the coefficient of friction is μ, then the maximum length ℓ_1 of the part of this rope which can overhang from the edge of the table without sliding down is

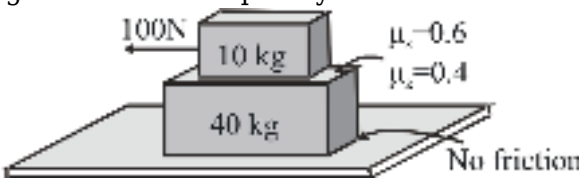
(1) $\frac{\ell}{\mu}$

(2) $\frac{\ell}{\mu + \ell}$

(3) $\frac{\mu\ell}{1 + \mu}$

(4) $\frac{\mu\ell}{\mu-1}$

44) A 40 kg slab rests on a frictionless floor. A 10 kg block rests on top of the slab. The static coefficient of friction between the block and the slab is 0.60 while the kinetic coefficient of friction is 0.40. The 10 kg block is acted upon by a horizontal force of 100 N. The resulting acceleration of the slab will be :-



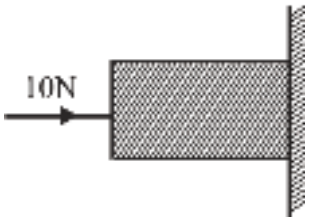
(1) 1.5 ms^{-2}

(2) 2.0 ms^{-2}

(3) 10 ms^{-2}

(4) 1.0 ms^{-2}

45) A horizontal force of 10 N is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2. The weight of the block is-



(1) 20 N

(2) 50 N

(3) 100 N

(4) 2 N

CHEMISTRY

1) What should be the density of an aqueous solution of urea (molar mass = 60 g/mol) such that the molality and molarity of the solution become equal to 1?

(1) 1.0 g/ml

(2) 1.6 g/ml

(3) 1.06 g/ml

(4) 1.16 g/ml

2) Equal volumes of 0.1 M $AgNO_3$ and 0.2 M NaCl are mixed. The concentration of NO_3^- ions in the mixture will be-

(1) 0.1 M

(2) 0.05 M

(3) 0.2 M

(4) 0.15 M

3) Volume strength of H_2O_2 solution represents :-

(1) Volume of H_2O_2 solution

(2) Moles of H_2O_2 solution

(3) Volume of O_2 in litre obtained from 1 litre H_2O_2 solution at STP

(4) Volume of H_2 gas obtained at NTP

4) 20 % Benzoic acid by weight dissolved in benzene then benzoic acid show 80% dimerisation in it. Calculate boiling point of solution ? [K_b of Benzene = 2.44 K.Kg. mol^{-1} , $T_b^o = 80^o$ C]

(1) 20°C

(2) 80°C

(3) 160°C

(4) 83°C

5) Match List - I with List - II.

List - I		**List - II**	
(a)	0.001M Na_2SO_4 and 0.003M glucose at same temperature	(i)	Hypotonic solution
(b)	0.003M Na_2SO_4 is _____ with 0.003M glucose at same temperature	(ii)	iCRT
(c)	0.001M Na_2SO_4 is _____ with 0.01M glucose at same temperature.	(iii)	Isotonic solution
(d)	Osmotic pressure	(iv)	Hypertonic solution

(1) (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)

(2) (a) - (i), (b) - (iii), (c) - (iv), (d) - (ii)

(3) (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)

(4) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)

6)

State T(true)/F(false) for following statements regarding given liquid solutions :



(a) Final volume of solutions if equal volume of compounds are mixed, is in the order:
$V_I = V_{II} < V_{III}$

(b) ΔS_{mix} for solution I and solution II is positive but for solution III it is zero.

(c) Enthalpy change of mixing is negative for solution II and solution III but positive for solution I.

(1) TFF

(2) FFF

(3) TTF

(4) FTT

7) Match the column.

	Column-I		Column-II
A	$20VH_2O_2$	P	2.5 M
B	24.5% w/v H_2SO_4	Q	1.76 M
C	Pure water	R	1.5 M
D	5% w/w NaOH ($d_{solution}$ = 1.2 gm/ml)	S	55.5 M

(1) A-R, B-Q, C-S, D-P

(2) A-S, B-P, C-R, D-Q

(3) A-P, B-S, C-Q, D-R

(4) A-Q, B-P, C-S, D-R

8) In a solution, if three molecules of solute undergoes association as : $3B \rightleftharpoons B_3$ then to evaluate degree of association 'α' the formula will be :-

(1) $\alpha = \frac{3(1-i)}{2}$

(2) $\alpha = \frac{i-1}{2}$

(3) $\alpha = \frac{3(i-1)}{2}$

(4) $\alpha = \frac{i-1}{3}$

9) The solubility of gases in liquids :

(1) Increases with increase in pressure and temperature

(2) Decreases with increase in pressure and temperature

(3) Increases with increase in pressure and decrease in temperature

(4) Decreases with increase in pressure and increase in temperature

10) According to Henry's law, the partial pressure of gas (P_g) is directly proportional to mole fraction of gas in liquid solution, $P_{gas} = K_H.X_{gas}$, where K_H is Henry's constant. Which is incorrect ?

(1) K_H is characteristic constant for a given gas-solvent system

(2) Higher is the value of K_H, lower is solubility of gas for a given partial pressure of gas

(3) K_H has temperature dependence

(4) K_H decreases with increase of temperature

11) K_H value for Ar(g), CO_2(g) HCHO(g) and CH_4(g) are 40.39, 1.67, 1.83×10^{-5} and 0.413

respectively arrange these gases in the order of their increasing solubility.

(1) HCHO < CH_4 < CO_2 < Ar

(2) HCHO < CO_2 < CH_4 < Ar

(3) Ar < CO_2 < CH_4 < HCHO

(4) Ar < CH_4 < CO_2 < HCHO

12) **Assertion :** One molar aqueous solution has generally higher concentration than one molal aqueous solution in dilute solution.
Reason : One molar solution contains less solvent than one molal aqueous solution in dilute aqueous solution.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but the Reason is True.

13) The vapour pressure of pure liquid solvent A is 10 torr and at same temperature 1g of non volatile B is added to the 20g of A, its vapour pressure is reduced to 9 torr. If the molecular mass of A is 200, calculate the molecular mass of B.

(1) 100

(2) 170

(3) 90

(4) 150

14) **Assertion :** On mixing of two liquids, the solutions becomes hot, it implies that it shows negative deviation from Raoult's Law.
Reason : Solutions which show negative deviations are accompained by increase in volume.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

15) **Assertion :** When NaCl is added to water, a depression in freezing point is observed.
Reason : The lowering of vapour pressure of a solution causes depression in the freezing point.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

16) **Assertion :** 0.1 m aqueous solution of glucose has higher elevation in the boiling point than 0.1 m aqueous solution of urea.
Reason : K_b of water for both 0.1 m aqueous solution of glucose and 0.1 m aqueous solution of urea has different values.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not the correct explanation of assertion.

(3) Assertion is correct but reason is incorrect.

(4) Both assertion and reason are incorrect.

17) **Assertion :-** Azeotropic mixtures are formed by both ideal and non-ideal solutions.
Reason :- Boiling point of an azeotropic mixture is either higher than both the components or lower than both the components.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is false but reason is true.

(4) Both assertion and reason are false.

18) **Statement-I :-** An increase in surface area increases the rate of evaporation.
Statement-II :- Stronger the intermolecular attractive forces, fast is the rate of evaporation at a given temperature.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) Both **Statement I** and **Statement II** are correct.

(3) **Statement I** is correct but **Statement II** is incorrect.

(4) **Statement I** is incorrect but **Statement II** is correct.

19) Which of the following solutions will have highest boiling point ?

(1) 1% w/w $CaCO_3$ in water

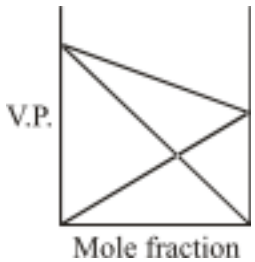
(2) 1% w/w NaCl in water

(3) 1% w/w $Ca_3(PO_4)_2$ in water

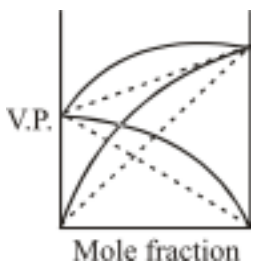
(4) 1% w/w Urea in water

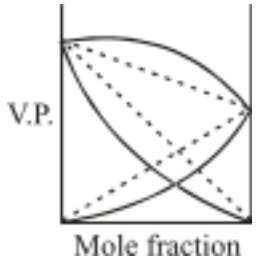
20) If C_2H_5OH and H_2O solution is example of non-ideal solution then which graphical representation is correct ?

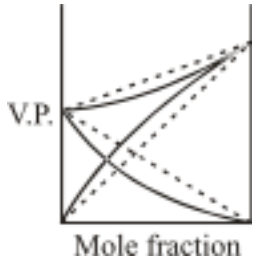
(1)



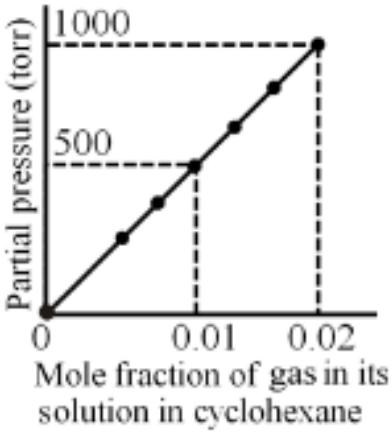
(2)



(3) 

(4) 

21) If we draw a graph between partial pressure of a gas versus mole fraction of the gas in solution, then we get following graph :-



The Henry's law constant, K_H is -

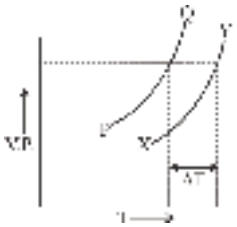
(1) 5×10^4 torr

(2) 10^4 torr

(3) 10^{-4} torr

(4) 2×10^{-5} torr

22) In the graph plotted between vapour pressure (V.P.) and temperature (T) :-



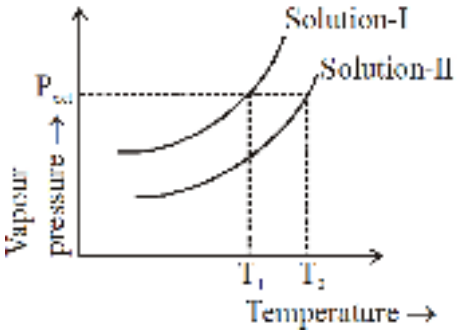
(1) PQ is the curve for solvent, XY is the curve of solution and ΔT is depression in freezing point

(2) PQ is the curve for solution, XY is the curve for solvent and ΔT is elevation in boiling point

(3) PQ is the curve for solvent, XY is the curve for solution and ΔT is molal depression in boiling point

(4) PQ is the curve for solvent, XY is the curve for solution and ΔT is elevation in boiling point.

23) Two different solutions were prepared by dissolving sucrose in water having concentration of 1m and 2m respectively. Value of $(T_2–T_1)$ will be:-



[K_b = 0.52 K Kg. m^{-1}]

(1) 1 K

(2) 1.86 K

(3) 0.52 K

(4) 1.86° C

24) **Assertion :–** Isotonic solutions don't show net phenomenon of osmosis.
Reason :– Isotonic solutions have equal osmotic pressure at constant temperature.

(1) Assertion and reason are true and reason is the correct explanation of assertion.

(2) Assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion and reason are false.

25) **Assertion :** Strong acid HNO_3 and water solution shows negative deviation from Raoult Law.
Reason : This solution forms maximum boiling Azeotrope.

(1) Assertion and reason are correct and reason is not the correct explanation of assertion

(2) Assertion is true and reason is false

(3) Assertion is false and reason is true

(4) Assertion and reason are correct and reason is the correct explanation of assertion

26) **Assertion :** The concentration of pollutants in water or atmosphere is often expressed in ppm.
Reason : Concentration in parts per million can be expressed as mass to mass, volume to volume and mass to volume.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

27) **Assertion :** For an ideal solution, ΔH_{mix} is zero.
Reason : In an ideal solution A-B interaction are less than A-A & B-B interactions.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

28) Assertion (A): Raoult's Law is a special case of Henry's Law.
Reason (R): The partial vapour pressure of a component is directly proportional to its mole fraction in solution.

(1) Both assertion and reason are false.
(2) Assertion is correct and reason is incorrect.
(3) Assertion is incorrect and reason is correct.
(4) Both assertion and reason are correct.

29) The Vant Hoff factor for $3A \rightarrow A_3$ is given by (100% Association):-

(1) 3
(2) $\frac{1}{3}$
(3) $\frac{1}{2}$
(4) $\frac{2}{3}$

30) Rectified aqueous spirit contains 92% ethanol by weight. The mole fraction of ethanol be :-

(1) 0.818
(2) 0.99
(3) 0.118
(4) 0.61

31) The normality of 2 molar H_3PO_2 solution will be :-

(1) 2 N
(2) 4 N
(3) 6 N
(4) 3 N

32) 6.8 g of H_2O_2 is dissolved in 1L of Solution volume strength of H_2O_2 is -

(1) 22.4
(2) 11.2
(3) 2.24
(4) 1.12

33) The vapour pressure of 10% mole aqueous glucose solution is: ($P^0_{H_2O}$ = 100 mm of Hg)

(1) 10
(2) 100
(3) 90
(4) 95

34) The molal elevation constant of water = 0.52 K kg mol^{-1}. The boiling point of 1.0 molal aqueous KCl solution should be :-

(1) 100.52°C

(2) 101.04°C

(3) 99.48°C

(4) 98.96°C

35) Relative decrease in vapour pressure of aqueous solution of NaCl is 0.18. Then number of moles of NaCl present in 180g of H_2O is :-

(1) 2.2 mol

(2) 1.1 mol

(3) 3.2 mol

(4) 4.2 mol

36) For a reaction; $A + B \rightleftharpoons C + D$, the initial concentration of A and B are equal but the equilibrium concentration of C is twice that of equilibrium concentration of A. The K_c is :

(1) 4

(2) 9

(3) 1/4

(4) 1/9

37) For which of the following reactions $K_p < K_c$:-

(1) $N_2O_3(g) \rightleftharpoons NO_2(g) + NO(g)$

(2) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(4) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

38) The equilibrium constants for the formation of oxides of nitrogen are given as under (at 25°C):
(a) $N_2 + O_2 \rightleftharpoons 2NO$; $K = 5 \times 10^{-31}$
(b) $N_2 + 2O_2 \rightleftharpoons 2NO_2$; $K = 8 \times 10^{-31}$
(c) $2N_2 + 5O_2 \rightleftharpoons 2N_2O_5$; $K = 1 \times 10^{-34}$
Choose the correct statement :-

(1) NO_2 is the least stable oxide

(2) N_2O_5 is the most stable oxide

(3) The stability order should be : $N_2O_5 > NO_2 > NO$

(4) The stability order should be : $NO_2 > NO > N_2O_5$

39) In a reversible reaction, two substances are in equilibrium. If the concentration of each one is doubled, the equilibrium constant will be :-

(1) Doubled

(2) Constant

(3) Halved

(4) Tripled

40) The equilibrium constant of a reaction is 20.0. At equilibrium, the rate constant of forward reaction is 10. The rate constant for backward reaction is :

(1) 0.5

(2) 2.0

(3) 10.0

(4) 200.0

41) 2 mole NaN_3 dissolved in 1000 g water and solution is cooled to –10°C then calculate weight of ice form.
Given K_f (H_2O) = 1.8 K.kg/mol

(1) 110 g

(2) 220 g

(3) 280 g

(4) 360 g

42) **Assertion (A) :** The boiling point of 0.1 M urea solution is less than that of 0.1 M KCl solution.
Reason (R) : Elevation of boiling point is directly proportional to the number of moles of non-volatile solute particles present in the solution.
Choose the correct answer from the options given below :

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false.

(3) A is false but R is true.

(4) Both A and R are true and R is the correct explanation of A.

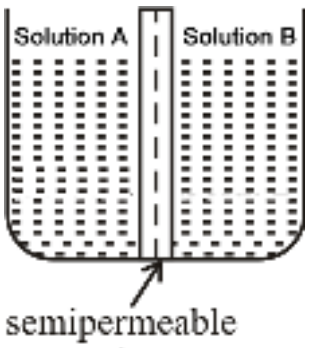
43) 6.02×10^{20} molecules of solid $CaCO_3$ are present in 100 ml of solution, then its active mass is :

(1) 0.01

(2) 1

(3) 0.001

(4) 0.1



44) In the above setup solvent flows from A to B then :

(1) Solution A is more dilute then solution B

(2) Solution A is more concentrated then solution B

(3) Solution A and solution B are equally concentrated

(4) Solution A is at higher temperature relative to solution B

45)

Consider the following cases :-
Case I : 2M CH_3COOH solution in benzene at 27°C where there is dimer formation to the extent of 100%.
Case II : 0.5 M KCl aqueous solution at 27°C, which ionises 100%

Which is correct ?

(1) Both are isotonic

(2) I is hypertonic

(3) II is hypertonic

(4) None is correct

BIOLOGY

1) **Assertion :-** S.L. Miller with his experiment proved Big-bang Theory.
Reason :- Big-bang Theory explain origin of Life.

(1) Both Assertion & Reason are correct & Reason is correct explanation of Assertion.

(2) Both Assertion & Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct & Reason is incorrect

(4) Both Assertion & Reason are incorrect

2) **Assertion :** An analysis of meteorite supports chemical evolution.
Reason : Analysis of meteorite content revealed similar compounds indicating that similar process are occurring elsewhere in space.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

3)

Fill in the blanks with correct option :
Oparin of ___A___ and Haldane of ___B___ proposed chemosynthetic theory.

(1) A - Russia, B - England

(2) A - England, B - Russia

(3) A - France, B - England

(4) A - America, B - Russia

4) Select the correct statement:

(1) Life originated on earth after 4 billion years of origin of earth

(2) Prokaryotic cells are formed after the formation of eukaryotic cells

(3) Water and oxygen are two essential components for the origin of life through chemosynthesis

(4) First non – cellular forms of life would have been giant molecules like RNA, Protein etc.

5) Identify the correct match from column I and column II :–

Column–I	Column–II
(A) Origin of Universe	(i) 4 bya
(B) Origin of earth	(ii) 4.5 bya
(C) Origin of first cellular form of life	(iii) 3 bya
(D) Origin of first non-cellular form of life	(iv) 13.8 bya
(E) Life appeared	(v) 2 bya

(1) A – iv, B – i, C – iii, D – ii, E – v

(2) A – iv, B – i, C – v, D – ii, E – iii

(3) A – iv, B – ii, C – v, D – iii, E – i

(4) A – iv, B – ii, C – iii, D – v, E – i

6) **Statement-I :** S.L. miller with his experiment proved the theory of biogenesis.
Statement-II : Theory of biogenesis attemps us to explain the origin of universe.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

7) **Statement I :-** The conditions or early earth were low temperature, volcanic storms, oxidising atmosphere containing CH_4, NH_3 ete.
Statement II :- The first non-cellular forms of life could have originated 3 billion years back.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but II is incorrect.

(4) Statement I is incorrect but II is correct.

8) Given below are two statements :-
Statement-I : The earth is approximately 4.5 billion years old.
Statement-II : Louis Pasteur's experiments disproved the theory of spontaneous generation.
In the light of above statements choose the correct answer from the options given below :-

(1) Only statement-I is correct

(2) Only statement-II is correct.

(3) Both statements are correct.

(4) Both statements are incorrect.

9) Statement-I : Miller observed formation of amino acids in his experiment.
Statement-II : Oparin and Haldane proposed that the first form of life could have come from pre-existing non-living organic molecules.

(1) Statement-I is correct but statement-II is incorrect.

(2) Statement-I is incorrect but statement-II is correct.

(3) Both statement-I & statement-II are correct.

(4) Both statement-I & statement-II are incorrect.

10) First organisms which inhabitated on earth were :-

(1) Chemoheterotrophs

(2) Chemoautotrophs

(3) Photoautotrophs

(4) Mixotrophs

11) Which type of atmosphere was present on earth during chemical evolution ?

(1) Reducing

(2) Presence of ozone layer

(3) Oxidising

(4) Absence of U.V. rays

12) Atmosphere of earth just before the origin of life consisted of :-

(1) Water vapours, CH_4 , NH_3 and oxygen

(2) O_2 , NH_3 and CH_4

(3) CH_4, NH_3 , H_2 and water vapours

(4) CH_4, O_3 , O_2 and water vapours

13) Analysis of, meteorite content and Miller's experiment proves :-

(1) Chemical evolution

(2) Biological evolution

(3) Organic evolution

(4) Co-evolution

14) The first cellular form of life originated before how many years :-

(1) 3 billion years before

(2) 4 billion years before

(3) 2000 million years before

(4) 0.5 million years before

15) Spontaneous generation theory was disapproved by :-

(1) S.L. Miller
(2) Greek Thinkers
(3) Louis Pasteur
(4) Oparin

16) According to early Greek thinkers units of life which were transferred to different planets including earth were called :-

(1) Gametes
(2) Factors
(3) Cells
(4) Spores

17) Earth and other planets formed from :-

(1) aggregates of uranium
(2) cloud of gas and dust
(3) division of pre existing planets
(4) collisions of meteroites

18) Which theory explains origin of universe ?

(1) Special creation theory
(2) Big-Bang theory
(3) Oparin-Haldane theory
(4) Abiogenesis

19) Which theory states that 'All living organisms that we see today were created as such' ?

(1) Cosmic panspermia theory
(2) Theory of special creation
(3) Theory of biogenesis
(4) Oparin-Haldane theory

20) Match the following :-

Column I		**Column II**	
(A)	Common cold	(i)	Dry & Scaly lesions
(B)	Typhoid	(ii)	*Haemophilus influenzae*
(C)	Pneumonia	(iii)	Stool with mucous & blood clots
(D)	Malaria	(iv)	Rhino virus

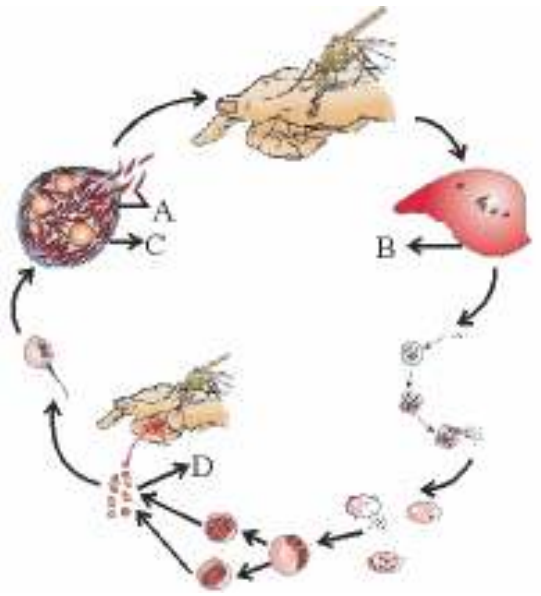
(E)	Ringworm	(v)	*Salmonella typhi*
(F)	Amoebiasis	(vi)	Haemozoin

(1) A–iv, B–v, C–ii, D–vi, E–i, F–iii

(2) A–ii, B–iv, C–i, D–iv, E–iii, F–v

(3) A–iv, B–v, C–ii, D–vi, E–iii, F–i

(4) A–iv, B–v, C–ii, D–i, E–vi, F–iii



21) Identify the A, B, C, D in the following life cycle.

Options :-

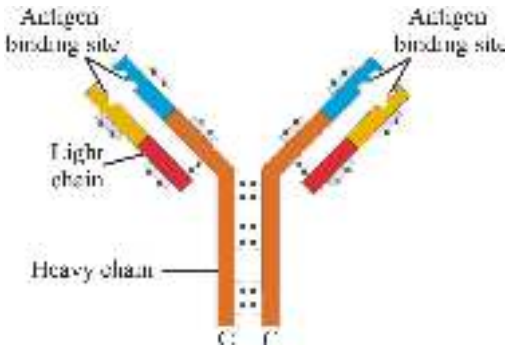
	A	B	C	D
(1)	Gametocyte	Salivary gland	Liver	Sporozoites
(2)	Sporozoites	Salivary gland	Liver	Gametocyte
(3)	Sporozoites	Liver	Salivary gland	Gametocyte
(4)	Sporozoites	Liver	Gametocyte	Salivary gland

(1) 1

(2) 2

(3) 3

(4) 4



22) How many of the below are correct about given diagram :-

(a) Gives cell mediated immunity
(b) T-cells themselves do not secrete but help
B-cell to produce them.
(c) Disulphide bond present
(d) H_2L_2 molecule
(e) Glyco protein molecule

(1) One
(2) Two
(3) Three
(4) Four

23) The following table shows certain diseases, their causative organisms and symptoms. Choose the option which correctly fills the gaps in the table :-

	Name of disease	**Causative organism**	**Symptoms**
	Ascariasis	*Ascaris*	_____(A)
	_____(B)	*Trichophyton*	Appearance of dry, scaly lesions on various parts of body
	_____(C)	Rhino virus	Nasal congestion and discharge, sore throat, cough, headache
	Filariasis	_____(D)	Inflammation of lower limbs

(1) A – Internal bleeding, muscular pain, fever, anemia and blockage of intestinal passage
C – Pneumonia

(2) B – Ringworm
D – *Wuchereria*

(3) A – High fever, stomach pain, constipation, headache and loss of appetite
C – Common cold

(4) B – Athlete's food
D – *Dracunculus medinensis*

24)

Identify the correct match from the columns I, II and III

Column-I	**Column-II**	**Column-III**
1. Bacteria	(a) *Plasmodium*	i. Infection of lungs

2. Virus	(b) *Wuchereria bancrofti*	ii. Fever with chill
3. Protozoa	(c) *Haemophilus influenzae*	iii. Inflammation of lower limbs
4. Helminth	(d) Rhino virus	iv. Infection of upper respiratory tract

(1) 2–c–i, 1–d–iv, 3–4–ii, 4–b–iii

(2) 3–a–ii, 4–b–iii, 2–c–iv, 1–d–i

(3) 4–b–iii, 3–a–ii, 1–c–i, 2–d–iv

(4) 4–b–iii, 3–a–i, 1–c–ii, 2–d–iv

25) **Assertion (A) :–** Everyday we are exposed to large number of infectious agents. However only few of these exposures result in disease.
Reason (R) :– Our body is able to defend itself from most of these foreign agents. This overall ability of host to fight the disease causing organism conferred by immune system is called immunity.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

26) **Assertion :-** Rheumatoid arthritis is an example of autoimmune disorder.
Reason :- In rheumatoid arthritis, IgM antibody is formed against synovial membrane.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

27) **Assertion :-** Coca alkaloid or cocaine is obtained from coca plant *Erythroxylum coca*, native to South America.
Reason :- Cocaine has a potent stimulating action on central nervous system.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

28) **Assertion :** Ionising radiations like X rays and gamma rays are example of carcinogens.
Reason : Agents which cause transformation of normal cells into cancerous cells are called carcinogens.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

29) **Assertion:** Filarial worm cause a slowly developing chronic inflammation of the organs.
Reason: Filarial worm is transmitted by female mosquito vector.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

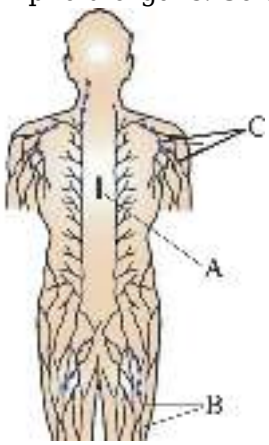
30) **Assertion:** AIDS is not a congenital disease.
Reason: AIDS is the deficiency of immune system, acquired during the lifetime of an individual.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

31) **Assertion :** Those who take drugs intravenously are much more likely to acquire serious infections like AIDS & hepatitis - B.
Reason : Both AIDS & hepatitis - B transmit by sharing infected needles.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

32) The figure below show diagrammatic representation of lymphoid organs. Select the correct option having label which shows primary lymphoid organ :-



(1) A
(2) B
(3) C
(4) A and B both



33) Identify the diagram shown below :-

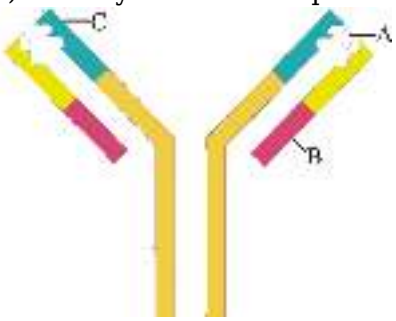
(1) *Cannabis Sativa*

(2) Datura

(3) Opium poppy

(4) *Erythroxylum*

34) Identify the correct option for given diagram for A, B and C :-



	A	B	C
(1)	Epitope	Light chain	Heavy chain
(2)	Paratope	Constant light chain	Variable heavy chain
(3)	Epitope	Constant light chain	Constant heavy chain
(4)	Paratope	Constant light chain	Constant heavy chain

(1) 1

(2) 2

(3) 3

(4) 4



35) Choose the correct option related to below shown human disease..... :-

	Disease	Tissue/organ affected
(1)	Elephantiasis	blood vessel
(2)	Ascariasis	stomach
(3)	Wuchereriasis	lymphatic vessels
(4)	Amoebiasis	liver

(1) 1

(2) 2

(3) 3

(4) 4



36) Which statement is correct about given diagram? (a) *Plasmodium* responsible for this.
(b) Allergic reactions.
(c) Caused by *Wuchereria bancrofti*
(d) Heat and moisture help these to grow

(1) a, b, d only

(2) c, d only

(3) d only

(4) All of these

37) Which of the following statement is false ?

(1) The adverse effects are just not restricted to the person who is using drug or alcohol.

(2) At times, a drug/alcohol addict becomes the cause of mental and financial distress to his/her entire family and friends.

(3) Excessive doses of drug may lead to coma and death due to respiratory failure, heart failure or cerebral haemorrhage.

(4) Withdrawal symptoms can never be severe or life threatening.

38) Which of the following statement is incorrect about amoebic dysentery ?

(1) It is caused by *Entamoeba histolytica*

(2) *Culex* mosquito act as a mechanical carrier

(3) It is a parasite of large intestine

(4) Patients suffer from constipation, Abdominal pain and cramps.

39) Which of the following options gives the correct categorisation of barriers of Innate immunity ?

	Physical	Physiological	Cellular
(1)	N.K. cell	Saliva	PMNL neutrophils

(2)	Skin	Tear	W.B.C.
(3)	Mucous	Interferon	T-killer cells
(4)	Fever	Phagocyte cells	Leukocyte

(1) 1

(2) 2

(3) 3

(4) 4

40) Read the following statements (A-D) :
A. Malignant malaria is caused by *Plasmodium vivax*.
B. Typhoid caused by *Haemophilus influenzae*.
C. Common cold is caused by Rhino virus.
D. Amoebiasis is caused by bacteria.
How many of the above statements are correct ?

(1) Four

(2) One

(3) Two

(4) Three

41)

Which of the following statements are correct ?
(a) Health is affected by genetic disorders, life style and infections.
(b) Balanced diet, personal hygiene and regular exercise are very important to maintain good health.
(c) Health is "absence of disease" or physical fitness" only.

(1) a, c

(2) b, c

(3) a, b, c

(4) a,b

42) In the list given below, how many are secondary lymphoid organs ?
Thymus, Spleen, Bone marrow, Tonsil, Appendix

(1) 2

(2) 3

(3) 5

(4) 4

43) Diseases are broadly grouped into infections and non infections diseases. How many of the following are infections diseases :-
(i) Small pox (ii) Allergy (iii) Influenza
(iv) AIDS (v) Cancer

(1) One

(2) Two

(3) Three

(4) Four

44)

Our health is affected by :-
(a) Genetic disorders
(b) Infections

(c) Life style

(1) a only

(2) b, c only

(3) a, b only

(4) a, b, c

45) Which disorder is related with smoking ? (a) Lung cancer
(b) Bronchitis
(c) Emphysema
(d) Coronary heart disease
(e) Gastric ulcer
(f) Urinary bladder cancer (g) Throat cancer

(1) a, b, e, g

(2) a, b, c, e, f

(3) e, d, f

(4) a, b, c, d, e, f, g

46) How many of the following diseases are caused by protozoa ? Filariasis, Common cold, Malaria, Ascariasis, Typhoid, Gonorrhoea

(1) Four

(2) Two

(3) Five

(4) One

47) **Statement-I :** In primary Lymphoid organs immature Lymphocytes differentiate into antigen-sensitive Lymphocytes.
Statement-II : Spleen is the main lymphoid organ where all blood cells including lymphocytes are produced.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II is incorrect.

48) **Statement-I :** T-lymphocytes mediate CMI.

Statement-II : CMI is responsible for the graft rejection.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

49) **Statement-I :** Normal cells does not show a property called contact inhibition.
Statement-II : Metastasis is the most feared property of benign tumors.

(1) Statement-I & Statement-II both are correct.
(2) Statement-I is correct & Statement-II is incorrect.
(3) Statement-I is incorrect & Statement-II is correct.
(4) Both Statement-I & Statement-II are incorrect.

50) Cross match the following column-I and Column-II :-

	Column I		**Column II**
(i)	Active natural	(a)	Antiserum
(ii)	Active artificial	(b)	Colostrum
(iii)	Passive natural	(c)	Vaccination
(iv)	Passive artificial	(d)	Infection

	i	**ii**	**iii**	**iv**
(1)	a	b	c	d
(2)	b	c	d	a
(3)	c	d	a	b
(4)	d	c	b	a

(1) 1
(2) 2
(3) 3
(4) 4

51)

(A)	IgG	(i)	Allergic & parasitic infections
(B)	IgM	(ii)	Ist antibody to appear in primary immune response
(C)	IgA	(iii)	Most abundant antibody
(D)	IgE	(iv)	Antibody present in saliva

(1) A-(iii); B-(ii); C-(iv); D-(i)
(2) A-(iii); B-(ii); C-(i); D-(iv)
(3) A-(ii); B-(iii); C-(iv); D-(i)
(4) A-(ii); B-(iv); C-(iii); D-(i)

52) Match the following-

Column I		Column II	
A	Chemical cancer causing agent	1	Cigarette smoke
B	Physical cancer causing agent	2	X-ray, gamma rays
C	Biological cancer causing agent	3	Oncogenic viruses

(1) A=3 B=2 C=1

(2) A=1 B=2 C=3

(3) A=2 B=1 C=3

(4) A=1 B=3 C=2

53) Which of the following options gives the correct categorisation of lymphoid organs with structure and type :-

	Column-I		Column-II		Column-III
A	Spleen	a	Lobed organ	i	Primary lymphoid organ
B	Lymph node	b	Bean shaped	ii	Large reservior of RBC
C	Thymus	c	Long bones	iii	Secondary lymphoid organ
D	Bone marrow	d	Solid	iv	Main lymphoid organ

(1) A-a-ii B-b-iii C-c-i D-d-iv

(2) A-b-ii B-d-iv C-a-i D-c-iii

(3) A-b-ii B-d-iii C-a-i D-c-iv

(4) A-c-iii B-d-iv C-c-i D-b-ii

54) Which of the following is not a vector borne disease ?

(1) Dengue

(2) Chikungunya

(3) Pneumonia

(4) Malaria

55) Find out **incorrect** match :-

(1) Allergy – Asthma

(2) Autoimmune disorder – Down syndrome

(3) Active immunity – Vaccination

(4) Passive immunity – IgA in colostrum

56) Lightest antibody is ?

(1) IgG
(2) IgM
(3) IgE
(4) IgD

57) Non-phagocytic cell is :-

(1) Neutrophil
(2) Monocyte
(3) Macrophage
(4) Natural Killer Cell

58) Amoebiasis is caused by ?

(1) *Shigella dysenterae*
(2) *Entamoeba hystolytica*
(3) *Ascaris lumbricoids*
(4) *Wuchereria*

59) Humoral immunity is mediated by :-

(1) Eosinophil
(2) Neutrophil
(3) Plasma cell
(4) Cytotoxic T-cell

60) Bean shaped large organ has the lymphocyte and phagocyte and is the reservoir for erythrocyte. The organ is :-

(1) Primary lymphoid organ
(2) Secondary lymphoid organ
(3) Lymph node
(4) Tonsil

61) Widal test is performed to confirm :-

(1) Malaria
(2) Pneumonia
(3) Jaundice
(4) Typhoid

62) Immunity acquired after an infection is :-

(1) Active immunity
(2) Passive immunity
(3) Innate immunity
(4) Both (2) and (3)

63) Gametocytes during life cycle of *Plasmodium* are developed in :-

(1) RBCs of mosquito
(2) Stomach of mosquito
(3) Liver of man
(4) RBCs of man

64) Which of the following is a viral disease :-

(1) Malaria
(2) Typhoid
(3) Hepatitis B
(4) Plague

65) Intestinal perforation and death may occur in severe cases of which disease ?

(1) Pneumonia
(2) Typhoid
(3) Filariasis
(4) Malaria

66) Blood circulation was discovered by :-

(1) Alec jaffrey
(2) William Harvey
(3) Thomas Addison
(4) Robert Brown

67) A form of malaria which is the most serious one and can even be fatal is caused by :

(1) *P.vivax*
(2) *P.malaria*
(3) *P.falciparum*
(4) *P.ovale*

68) The toxic substance, 'haemozoin', related to the high fever and chill, is released during which of the following disease ?

(1) Dengue
(2) Malaria
(3) Diphtheria

(4) Pneumonia

69) **Statement-I :** *Haemophilus influenzae* bacteria is responsible for the disease Pneumonia.
Statement-II : Pneumonia infects the alveoli of the lungs leading to severe problem in respiration.

(1) Statement-I & Statement-II both are correct.

(2) Statement-I is correct & Statement-II is incorrect.

(3) Statement-I is incorrect & Statement-II is correct.

(4) Both Statement-I & Statement-II are incorrect.

70) *Salmonella typhi* is a pathogenic bacterium which causes typhoid fever in human beings. These pathogens generally enter the small intestine through food and water contaminated with them and migrate to other organs through blood.
Given below are two statements about typhoid fever :
Statement-I : Sustained high fever (39°C to 40°C), weakness, stomach pain, constipation, headache and loss of appetite are some of the common symptoms of this disease.
Statement-II : Typhoid fever could be confirmed by Widal test.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

71) Read the following statements and find out the correct set of statements :
(A) The primary lymphoid organs are bone marrow and thymus where immature lymphocytes differentiate into antigen-sensitive lymphocytes.
(B) After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix.
(C) The thymus is quite large at the time of birth but keeps reducing in size with age and by the time puberty is attained it reduces to a very small size.
(D) The spleen is a large bean-shaped organ. It mainly contains lymphocytes and phagocytes.
(E) Lymph nodes serve to trap the micro-organisms or other antigens, which happen to get into the lymph and tissue fluid.
Options :

(1) A, B, C and E only

(2) B, C, D and E only

(3) A, B, C and D only

(4) A, B, C, D and E

72) Match the column-I and II and select the correct answer from the options :

	Column-I		**Column-II**
(A)	Physical barriers	(i)	PMNL-neutrophils

(B)	Physiological barriers	(ii)	Interferons
(C)	Cellular barriers	(iii)	Mucus coating of respiratory tract
(D)	Cytokines barriers	(iv)	Saliva in mouth

(1) A-i ; B-ii ; C-iii ; D-iv

(2) A-ii ; B-i ; C-iv ; D-iii

(3) A-iii ; B-iv ; C-i ; D-ii

(4) A-iv ; B-iii ; C-i ; D-ii

73) Given below are two statements : one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A) : When our body encounters a pathogen for the first time it produces a response called primary response which is of low intensity.
Reason (R) : Subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(2) Both **A** and **R** are true but **R** is not the correct explanation of **A**.

(3) **A** is true but **R** is false

(4) **A** is false but **R** is true

74) Given below are two statements :
Statement-I : The B-cells themselves do not secrete antibodies but help T-cells to produce them.
Statement-II : Tissue matching, blood group matching are essential before undertaking any graft/transplant.
In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Statement I is true but Statement II is false.

(2) Statement I is false but Statement II is true.

(3) Both Statement I and Statement II are true.

(4) Statement I is true but Statement II is false.

75) Which class of antibodies is primarily involved in the allergy ?

(1) IgG

(2) IgM

(3) IgA

(4) IgE

76) Which of the following correctly matches the technique with its principle ?

(1) CT scan– Uses strong magnetic fields and non-ionising radiation

(2) MRI – Provides three-dimensional X-ray images of internal organs
(3) Biopsy – Histopathological examination of thin tissue sections
(4) Widal test – Detection of cancer-specific antigens

77) Cocaine, obtained from *Erythroxylum coca*, mainly acts by :

(1) Inhibiting serotonin release
(2) Blocking dopamine transport
(3) Stimulating acetylcholine receptors
(4) Enhancing GABA activity

78) Pneumonia spreads mainly through :

(1) Contaminated food and water
(2) Droplets/aerosols from an infected person
(3) Insect bite
(4) Direct skin contact

79) Which bacteria are mainly responsible for causing pneumonia in humans by infecting the alveoli of the lungs ?

(1) *Streptococcus pneumoniae* and *Haemophilus influenzae*
(2) *Mycobacterium tuberculosis and Salmonella typhi*
(3) *Vibrio cholerae* and *Clostridium tetani*
(4) *Escherichia coli* and *Mycobacterium tuberculosis*

80) Elephantiasis, marked by swelling of lower limbs, is caused by :

(1) *Ascaris lumbricoides* – contaminated food and water
(2) *Wuchereria bancrofti* – female mosquito bite
(3) *Trichophyton* – skin contact with towels
(4) *Microsporum* – contaminated soil

81) Colostrum secreted during the initial days of lactation provides :

(1) IgM antibodies for active immunity
(2) IgE antibodies for allergic protection
(3) IgA antibodies for passive immunity
(4) IgG antibodies for autoimmune response

82) Which one is a wrong statement about HIV/AIDS ?

(1) It has no cure, prevention is best
(2) Infection may occur during blood transfusion
(3) It spreads through mosquito bites
(4) Ignorance can cause infection spread

83) Malignant tumors differ from benign tumors because malignant tumors :

(1) Are confined to one place
(2) Spread to other parts through blood (metastasis)
(3) Grow very slowly
(4) Do not damage surrounding tissues

84) Morphine, the precursor of heroin, is extracted from :

(1) Leaves of *Cannabis sativa*
(2) Flower of *Datura stramonium*
(3) Latex of *Papaver somniferum*
(4) Roots of *Rauwolfia serpentina*

85) Read the following statements and find out the correct set of statements :
(A) AIDS stands for acquired Immuno Deficiency Syndrome, meaning of deficiency of immune system acquired during one's lifetime.
(B) The disease is caused by Human Immunodeficiency Virus, which is a member of a group of viruses called retroviruses.
(C) HIV spreads only through sexual contact and can never be transmitted through transfusion of blood or infected needles.
(D) The time lag between infection and the appearance of symptoms of AIDS may vary from a few months to many years.
(E) The diagnostic test widely used for determining AIDS is the enzyme linked immunosorbent assay.
Options :

(1) A, B, C and D only
(2) A, B, D and E only
(3) B, C, D and E only
(4) A, B, C, D and E only

86) Read the following statements and find out the correct set of statements :
(A) In our body, cell growth and differentiation is highly controlled and regulated, but in cancer cells, there is breakdown of these regulatory mechanisms.
(B) Normal cells show a property called contact inhibition by virtue of which contact with other cells inhibits their uncontrolled growth.
(C) Cancerous cells continue to divide for a limited number of times, forming large masses of tissues called tumors.
(D) Tumors are of two types : benign and malignant.
(E) Benign tumors normally remain confined to their original location and cause little damage.
Options :

(1) A, B, C and D only
(2) B, C, D and E only
(3) A, B, D and E only
(4) A, B, C, D and E

87) Read the following statements and find out the correct set of statements :

(A) Opioids are the drugs which bind to specific opioid receptors present in our central nervous system and gastrointestinal tract.
(B) Heroin, commonly called smack, is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound.
(C) Morphine is extracted from the latex of the poppy plant papaver somniferum.
(D) Opioids are generally taken by snorting and oral ingestion and they act as stimulants to increase body functions.
(E) Morphine is a very effective sedative and painkiller, which is very useful to patients who have undergone surgery.
Options :

(1) A, B, C and E only

(2) B, C, D and E only

(3) A, C, D and E only

(4) A, B, C, D and E

88) Read the following statements and find out the correct set of statements :
(A) Typhoid fever in human beings is caused by a pathogenic bacterium *Salmonella typhi*.
(B) These pathogens generally enters the small intestine through contaminated food and water and migrate to other organs through blood.
(C) Sustained high fever, weakness, stomach pain, constipation, headache, and loss of appetite are some of the common symptoms.
(D) In severe cases, intestinal perforation and death may occur in typhoid patients.
(E) Typhoid fever cannot be confirmed by the Widal test under any circumstances.
Options :

(1) A, B, C and D only

(2) B, C, D and E only

(3) A, C, D and E only

(4) A, B, C, D and E

89) Read the following statements and find out the correct set of statements :
(A) Allergy is the exaggerated response of the immune system of certain antigens present in the environment.
(B) The antibodies produced to these type of response are of IgE type.
(C) Common examples of allergens include mites in dust, pollens, animal dander, etc.
(D) Use of drugs like antihistamine, adrenalin and steroids quickly reduce the symptoms of allergy.
(E) Modern-day lifestyle has resulted in lowering of immunity and increased sensitivity to allergens, causing fewer people to suffer from allergies globally.
Options :

(1) A, B, C and D only

(2) B, C, D and E only

(3) A, C, D and E only

(4) A, B, C, D and E

90) Read the following statements and fill in the blanks with the correct option :
(A) Typhoid fever in humans is caused by the pathogenic bacterium _____ (i) _____.
(B) The common cold is an infectious human ailment caused by _____ (ii) _____.

(C) Chemically, heroin (smack) is _____ (iii) _____ which is obtained by acetylation of morphine.
(D) The diagnostic test widely used for detecting AIDS is _____ (iv) _____.
Options :

(1) (i) *Salmonella typhi*, (ii) Rhinoviruses, (iii) Diactylmorphine, (iv) ELISA

(2) (i) *Streptococcus pneumoniae*, (ii) Coronaviruses, (iii) Morphine, (iv) Widal test

(3) (i) *Salmonella typhi*, (ii) Retroviruses, (iii) Cannabinoid, (iv) PCR test

(4) (i) *Plasmodium vivax*, (ii) Rhinoviruses, (iii) Diactylmorphine, (iv) Western blot

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	4	3	1	4	1	4	2	4	2	1	4	4	1	1	1	1	3	3	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	1	1	1	1	3	3	3	4	1	3	4	2	4	2	2	4	2	4
Q.	41	42	43	44	45															
A.	2	3	3	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	3	2	3	4	3	2	4	1	3	4	3	1	3	3	1	4	3	3	2	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	3	1	1	2	3	4	2	1	1	3	3	2	2	1	2	4	2	1
Q.	86	87	88	89	90															
A.	3	4	2	1	1															

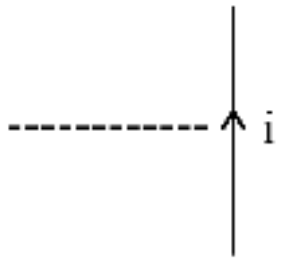
BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	1	4	3	4	4	3	3	1	1	3	1	3	3	4	2	2	2	1
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	2	3	1	1	2	1	2	1	1	1	2	2	3	3	4	2	2	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	2	3	4	4	4	2	1	4	4	1	2	3	3	2	1	4	2	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	1	4	3	2	2	3	2	1	3	4	3	2	2	4	3	2	2	1	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	3	2	3	2	3	1	1	1	1										

SOLUTIONS

PHYSICS

1)



Thum –i
Flat finger –r
Palm $-\vec{B}$
So $\vec{B}$ is out of plane

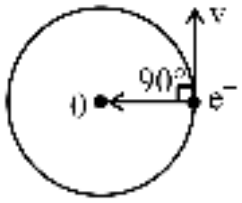
2)

Net magnetic field at O :

$B = B_+ + B_- = \frac{\mu_0 qv}{8\pi r^2} + \frac{\mu_0 qv}{4\pi r^2} = \frac{3\mu_0 qv}{8\pi r^2}$

Direction : into the plane $(\otimes)$.

3)

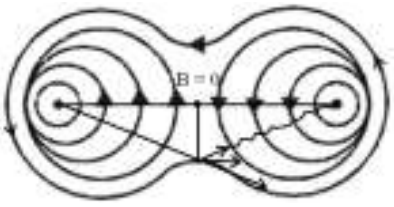


$B = \frac{\mu_0}{4\pi}\frac{qv\sin 90^\circ}{r^2}$

$B = \frac{\mu_0 qv}{4\pi r^2}$

$r^2 \propto \frac{v}{B} \Rightarrow r \propto \sqrt{\frac{v}{B}}$

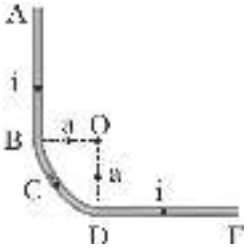
4)

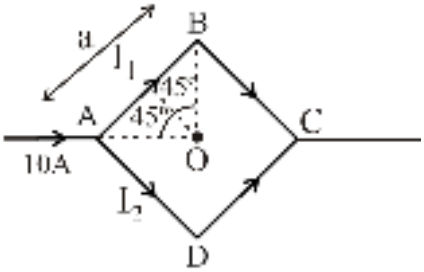


5) The field at O due to AB is $\frac{\mu_0}{4\pi}.\frac{i}{a}\hat{k}$ and that due to DE is also $\frac{\mu_0}{4\pi}.\frac{i}{a}\hat{k}$.

However the field due to BCD is $\frac{\mu_0}{4\pi}.\frac{i}{a}\left(\frac{\pi}{2}\right)\hat{k}$.

Thus the total field at O is $\frac{\mu_0}{4\pi}.\frac{i}{a}\left(2+\frac{\pi}{2}\right)\hat{k}$.





6)

$$I_1 = \frac{10 \times 2r}{2r + r} = \frac{20}{3}A$$

$$I_2 = \frac{10}{3}A$$

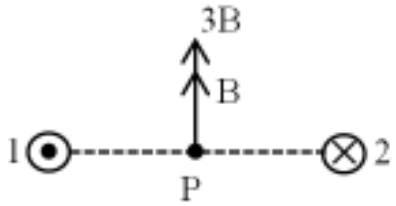
$$\vec{B}_{AB} = \vec{B}_{BC} = \frac{\mu_0 I_1}{4\pi\frac{a}{2}} \times 2\sin 45 \otimes$$

$$= \frac{\mu_0 I_1}{\sqrt{2}\pi a} \otimes$$

Similarily : $\vec{B}_{AD} = \vec{B}_{DC} = \frac{\mu_0 I_2}{\sqrt{2}\pi a} \odot$

$$B_{Net} = \frac{2\mu_0}{\sqrt{2}\pi a}(I_1 - I_2) \otimes$$

$$= \frac{10\sqrt{2}\mu_0}{3\pi a} \otimes$$



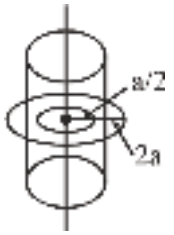
7)

($B_P = 4B$)

8)

Consider two amperian loops of radius a/2 and 2a as shown in the diagram. Applying Ampere's circuital law for these loops, we get

$$\oint B.dL = \mu_0 I_{enclosed}$$



For the smaller loop,

$$\Rightarrow B \times 2\pi\frac{a}{2} = \mu_0 \times \frac{I}{\pi a^2} \times \pi\left(\frac{a}{2}\right)^2 = \mu_0 I \times \frac{1}{4} = \frac{\mu_0 I}{4}$$

$\Rightarrow B' = \frac{\mu_0 I}{4\pi a}$, at distance $\frac{a}{2}$ from the axis of the wire.

Similarly, for bigger amperian loop

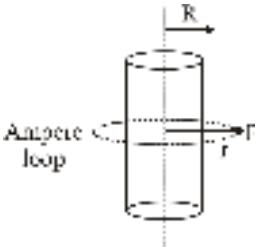
$B' \times 2\pi(2a) = \mu_0 I$

(total current enclosed by Amperian loop is 2.)

$\Rightarrow B' = \frac{\mu_0 I}{4\pi a}$

at distance 2a from the axis of the wire.

So, ratio of $\frac{B}{B'} = \frac{\mu_0 I}{4\pi a} \times \frac{4\pi a}{\mu_0 I} = 1$

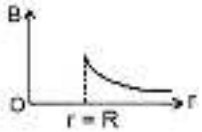


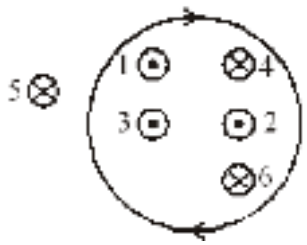
9)

(1) r < R, $B_p = 0$

(2) r ≥ R, $B_p = \frac{\mu_0 I}{2\pi r}$

$B_p \propto \frac{1}{r}$





10)

$\oint \vec{B}.\vec{d\ell} = \mu_0 \Sigma i$

$\oint B d\ell \cos 0^\circ = \mu_0[(1+5)-(1+2+4)]$

$= \mu_0[6-7]$

$$\oint B d\ell = -\mu_0 \text{Wb/m}$$

11)

Theoritical

12) $B = \mu_r \mu_o ni$
$= 4000 \times 4\pi \times 10^{-7} \times 1000 \times 5$
$= 8\pi$
$B = 25.12$ T

13) Mangetic force on positive and negative particles will be opposite direction.

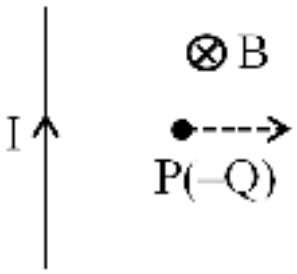
14) Conceptual

15)

Speed of charge moving in magnetic field constant
So $KE = \frac{1}{2}mv^2 = \text{constant}$

16) $T_e = \frac{2\pi m_e}{eB} = \frac{2\pi \times 9 \times 10^{-31}}{1.6 \times 10^{-19} \times 1.0 \times 10^{-4}}$

17) $\vec{F}_B = q\left(\vec{V} \times \vec{B}\right)$



18)
$F = -Q[\hat{i} \times -\hat{k}]$
$F = +[\hat{i} \times \hat{k}]$
$= -\hat{j}$
= –Y direction
Opposite to OY direction

19) **Explanation the Question:**
Proton has velocity with a component parallel to the field (along Y) and a perpendicular component; perpendicular part makes uniform circular motion, so overall path is a helix.

Concept: This question is based on Radius form $r = \frac{mv_\perp}{qB}$. Period from T = $\frac{2\pi m}{qB}$. Parallel velocity gives axial advance.

Formula:

$v_x = v \cos 60 = 1.0 \times 10^6$ m/s (perpendicular to B)

$v_{||} = v \sin 60 = 1.732 \times 10^6$ m/s (parallel to B)

$r = \frac{mv_\perp}{qB}$

$T = \frac{2\pi m}{qB}$

Solution:

$r = \frac{1.67 \times 10^{-27} \times 1.0 \times 10^6}{1.6 \times 10^{-19} \times 0.104} \approx 0.10m$

$T = \frac{2\pi \times 1.67 \times 10^{-27}}{1.6 \times 10^{-19} \times 0.104} \approx 2\pi \times 10^{-7}s.$

Final Answer : Option (3)

20) $p = \frac{2\pi m}{Bq}(v\cos 45^\circ) = \frac{2\pi m}{Bq}(v\sin 45^\circ)$

$\Rightarrow \frac{mv\sin 45^\circ}{Bq} = \frac{p}{2\pi}$

= radius of helix

21) $r = \frac{\sqrt{2mE_K}}{qB}$, $r \propto \frac{\sqrt{m}}{q}$

$q \propto C \propto \frac{1}{r}$ $\Rightarrow q \propto \frac{1}{r}$

$rH+ : r_{He+} : r_{0++} = \frac{\sqrt{1amu}}{e} : \frac{\sqrt{4amu}}{e} : \frac{\sqrt{16amu}}{2e}$

= 1 : 2 : 2

Radius of H^+ is smallest. So it will deflect maximum but deflection of He+ and O^{++} is equal.

22)

Radius of helix is

$r = \frac{mv\sin\theta}{q_0 B}$

$\Rightarrow v_\perp = v\sin\theta$

23) For case (a)

Acceleration = $\frac{m_2 g - F}{m_1 + m_2} = 3m/s^2$

for m_2, $m_2 g - T = m_2 a$

$\Rightarrow T = m_2(g - a) = 56$ N

For case (b)

Acceleration = $\frac{m_2g - m_1g - F}{m_1 + m_2} = 1 m/s^2$

For m_2, m_2g □ T = m_2a

⇒ T = $m_2(g$ □ a) = 72 N

24) $\frac{mg}{m(g-a)} = \frac{3}{2} \Rightarrow a = \frac{g}{3}$

25) $a_1 = \frac{F}{m_1}\ a_2 = \frac{F}{m_2}$

$a = \frac{F}{m_1 + m_2} = \frac{F}{\frac{F}{a_1} + \frac{F}{a_2}} = \frac{a_1 a_2}{a_1 + a_2}$

$a = \frac{3 \times 6}{3 + 6} = 2 m/s^2$

26)

Impulse = $m\ (\vec{v}_f - \vec{v}_i)$

$-3m\hat{i} = m\ (\vec{v}_f - \vec{v}_i)$

$\vec{v}_f = -3\hat{i} + \vec{v}_i$

$= -3\hat{i} + (2\hat{i} + 3\hat{j})$

$= \left(-\hat{i} + 3\hat{j}\right)\ m/sec.$

27)

Reaction on 6th coin due to fifth coin = weight of the coins above 6th coin

= 4 mg

28)

Since $T_C \cos 45° = T_B \sin 45° \Rightarrow T_C = T_B$ ...(1)

$T_C \sin 45° + T_B \cos 45° = 20g \Rightarrow 2T_C \times \frac{1}{\sqrt{2}} = 200$

$T_C = 100\sqrt{2}$ N = 141 N

29) Tension is the rope at every point will be F as acceleration of rope is zero.

30) $a_x = \frac{F \cos 60°}{M} = \frac{F}{2 \times 10} = \frac{F}{20}$

Thus $3 = (2 + 1) \times \frac{F}{20}$ □ F = 20 N.

31) When person dips finger in the water, water will apply upthrust & finger will apply same

force on water in downward direction so Net force in downward direction on water increases i.e. reading of balance increases.
Hence option (1)

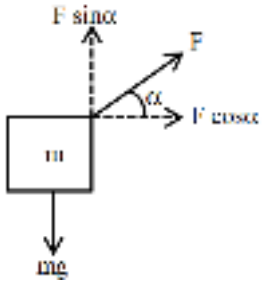
32)

Explain Question:
To calculate frictional force acting on the block.

Concept:
Friction

Solution:
Since the applied force is given as exactly half of this limiting value, so the block does not move.
F is insufficient to move the block.
Block is in equilibrium, therefore f = F.

Final Answer: Option (3)



33)
when body leaves the surface
F sin α = mg
Kt sin α = mg

$$t = \frac{mg}{K \sin \alpha}$$

34) The acceleration of a rocket is given by,

$$a = \frac{v}{m}\left(\frac{\Delta m}{\Delta t}\right) - g = \frac{400}{100}\left(\frac{5}{1}\right) - 10$$

$= (20 - 10) = 10\ m/s^2$

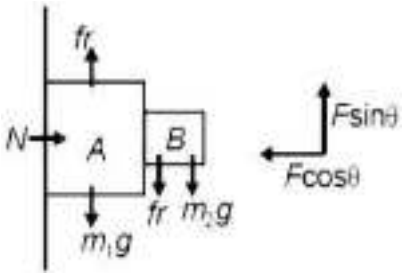
35) $\frac{dm}{dt}$ = Mass of bullet (m_B) × Bullets fired per sec (N).

Maximum force that man can exert F = $u\left(\frac{dm}{dt}\right)$.

$\therefore$ F = u × m_B × N

$$\Rightarrow N = \frac{F}{m_B \times u} = \frac{144}{40 \times 10^{-3} \times 1200} = 3$$

36)



From the free body diagram block 'A' has tendency to move downwards relative to block 'B'.
So, friction on block 'B' downwards.

37) **Explain Question :**
Calculating horizontal acceleration of a wedge for free fall.

Concept:
Pseudo force and relative acceleration.

Solution:
For the block to fall freely, the normal force from the wedge must be zero.

$$N + ma\sin\theta = mg\cos\theta$$

$$a = g \times \frac{\cos\theta}{\sin\theta} = \frac{g}{\tan\theta}$$

Final Answer: Option (2)

38)

$V^2 = (2g \sin\theta)S$ .....(1)

$$\left(\frac{V}{2}\right)^2 = 2[g\sin\theta - \mu g\cos\theta]S \quad .....(2)$$

$$\frac{1}{4} \times (2g\sin\theta)S = 2g[\sin\theta - \mu\cos\theta]S$$

$$\frac{\sin\theta}{4} = \sin\theta - \mu\cos\theta$$

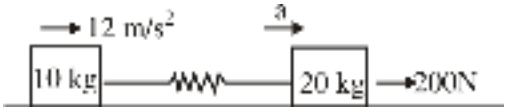
$$\mu\cos\theta = \sin\theta - \frac{\sin\theta}{4}$$

$$\mu\cos\theta = \frac{3\sin\theta}{4}$$

$$\mu = \frac{3\tan\theta}{4}$$

39)

$T = 10 \times 12 = 120$

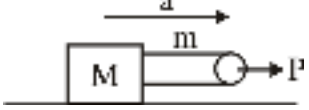


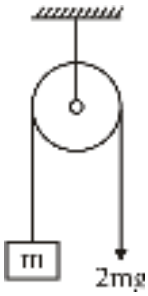
$200 - T = 20\ a$

$\boxed{a = 4 m/s^2}$

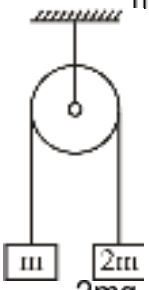
40) $a = \frac{P}{M+m}$

$T = Ma = \frac{MP}{M+m}$

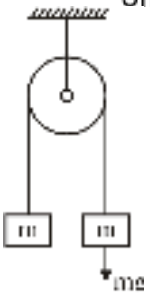


41) 

$a_1 = \frac{2mg - mg}{m} = g$



$a_2 = \frac{2mg - mg}{3m} = \frac{g}{3}$



$a_1 > a_3 > a_2$

$a_3 = \frac{2mg - mg}{2m} = \frac{g}{2}$

42)

$g_{eff} = g + a_0$

Acceleration of m_1 w.r.t. pulley

$= \frac{m_2 - m_1}{m_1 + m_2} g_{eff}$

$= \frac{m_2 - m_1}{m_1 + m_2}(g + a_0)$

43) For given condition we can apply direct formula $\ell_1 = \left(\frac{\mu}{\mu + 1}\right)\ell$

44) Common Acceleration :

$$a = \frac{100}{40+10} = 2\ m/s^2$$

40 kg : → $f_r = 40a = 80N$

∴ $f_{lim} = \mu_s = N = 0.6 \times 10\ g = 60N$

$f_{lim} = < f_r$ so move separately

so, Acceleration of 40kg slab is

$f_k = 40a$

$\mu_k N = 40a$

$0.4\ (10g) = 40a$

$a = 1\ m/s^2$

45) $mg = \mu N = \mu(10) = 2$

CHEMISTRY

46) **Explain the Question :** Find out density at which molality and molarity are equal.

Concept : This question is based on interconversion of con. terms.

Solution : Let molarity and molality of urea solution is 1.

$$M = \frac{\text{moles of solute}}{\text{vol. of solution (L)}}$$

$$m = \frac{\text{moles of solute}}{\text{vol. of solvent (kg)}}$$

if Molality (m) = 1 mol/kg → 1 mol urea in 1 kg solvent (water).

Molarity (M) = 1 mol/L → 1 mol in 1 L solution → volume of solution = 1000 mL

mass of solvent = 1 kg = 1000g

mass of solute = 1 × 60 = 60g

mass of solution = 1000 + 60 = 1060g

$$\frac{\text{mass}}{\text{vol.}} = d \Rightarrow d = \frac{1060}{1000} = 1.060 g/mL$$

Final answer : Option (3)

47) **Explain the Question :** We need the concentration of nitrate ions after mixing equal volumes of 0.1 M $AgNO_3$ and 0.2 M NaCl.

Concept : This question is based on This question is based on dilution effect and limiting reagent concept in precipitation reaction.

Solution : Reaction : $AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$

Before mixing : $[AgNO_3] = 0.1M$, $[NaCl] = 0.2\ M$

After mixing equal volumes, concentrations become half :

$[AgNO_3] = 0.05M$, $[NaCl] = 0.10M$

Ag^+ (0.05M) reacts completely with Cl^- (0.05 M of the 0.10M available).

Thus, 0.05 M $NaNO_3$ is formed, giving 0.05 M NO_3^-.

Final answer : 2

48) **Question Explanation:**

We are asked to determine what the volume strength of a hydrogen peroxide (H_2O_2) solution represents.

Volume strength is a term used to describe the concentration of hydrogen peroxide in solution.

Given:
The concept of volume strength is related to the amount of oxygen (O_2) released when hydrogen peroxide decomposes.

Concept:
Volume strength of H_2O_2 refers to the **volume of oxygen gas (O_2) produced at STP** by the decomposition of 1 liter of hydrogen peroxide solution.
The decomposition reaction of hydrogen peroxide is:
$2H_2O_2 \rightarrow 2H_2O + O_2$
From this, for every 2 moles of H_2O_2, 1 mole of O_2 is produced.

Calculation:
The volume strength indicates how much oxygen is released at standard conditions (STP) from a given volume of H_2O_2.

Final Answer:
Volume of O_2 in liter obtained from 1 liter H_2O_2 solution at STP (Option 3).

49) $\Delta T_b = T_b - T_b^\circ = iK_b \times m$

$$T_b - 80 = \left[1 + \left(\frac{1}{n} - 1\right)\beta\right] \times 2.44 \times \frac{20 \times 1000}{122 \times 80}$$

$$T_b - 80 = \left[1 + \left(\frac{1}{2} - 1\right) \times 0.8\right] \times 5$$

$= 0.6 \times 5$

$T_b = 83°C$

50)

Explain the question : Match solutions based on osmotic pressure comparison (hypertonic, hypotonic, isotonic) and formula of osmotic pressure.
Concept : This question is based on osmotic pressure $\pi = iCRT$.
• Hypertonic → higher π
• Hypotonic → lower π
• Isotonic → equal π
Solution :
(a) 0.001 M Na_2SO_4 vs 0.003 M glucose
• Na_2SO_4 → i = 3 → effective concentration = 0.001 × 3 = 0.003
• Glucose → i = 1 → 0.003 × 1 = 0.003
→ Equal, so isotonic → (iii)
(b) 0.003 M Na_2SO_4 vs 0.003 M glucose
• Na_2SO_4 → 0.003 × 3 = 0.009
• Glucose → 0.003 × 1 = 0.003
→ Na_2SO_4 has higher π → hypertonic → (iv)
(c) 0.001 M Na_2SO_4 vs 0.01 M glucose
• Na_2SO_4 → 0.001 × 3 = 0.003
• Glucose → 0.01 × 1 = 0.01

→ Na_2SO_4 has lower π → hypotonic → (i)
(d) Osmotic pressure formula → iCRT → (ii)
Final Answer : 3

51)

CCl_4 + Benzene ⇒ Positive deviation
Chloroform + acetone ⇒ Negative deviation
n-hexane + n-heptane ⇒ No deviation

52) **Question explanation :** Match each solution description to its molarity.
Concept : This question is based on Use standard conversions: volume strength, % w/v, % w/w with density, and pure water molarity.
Solution :

(A) V.S. = M × 11.2

20 = M × 11.2
M = 1.76

(B) $M = \frac{\%w/v \times 10}{M_{w,\,solute}} = \frac{24.5 \times 10}{98} = 2.5$

(C) for pure water

$M = \frac{1000}{18} = 55.5$

(D) $M = \frac{\%w/w \times d_{solution} \times 10}{M_{w,solute}}$

$= \frac{5 \times 1.2 \times 10}{40} = 1.5$

Final answer 4_ A-Q, B-P, C-S, D-R

53) **Explain question:** To find out α for $3B \rightleftharpoons B_2$

Concept: Van't Hoff factor.

Solution:
$3B \rightleftharpoons B_2$.

For association $i = 1 + \left(\frac{1}{n} - 1\right)\alpha$

for trimerisation n = 3

$i = 1 + \left(\frac{1}{3} - 1\right)\alpha$

$i = 1 - \frac{2}{3}\alpha$

$\Rightarrow \frac{2}{3}\alpha = 1 - i$

$\alpha = \frac{3(1-i)}{2}$

Final answer: Option (1)

54) **Question Explanation:** We need to find effect of pressure and Temperature on solubility of gas in liquid.

Concept: Solubility of gas in liquid.

Solution:
For Gas – liquid solution:-
P↑ ⇒ solubility ↑ (solubility ∝ P)
T↓ ⇒ solubility ↑ $\left(\text{Solubility} \propto \frac{1}{T}\right)$

Final answer: Option (3)

55)

NCERT (2019) Page No. # 41(Theory)

56)

Question Explanation:
We need to find correct order of solubility of gases using given values of K_H for given gases.
Concept:
Henry's law and solubility of gases.
Solution:
$P_{gas} = K_H X_{gas}$
$(X)_{gas} = \frac{P_{gas}}{K_H} \Rightarrow X_{gas} \alpha \frac{1}{K_H}$
⇒ Solubility of gas $\propto \frac{1}{K_H}$
Order of K_H ⇒
Ar > CO_2 > CH_4 > HCHO
(40.39) (1.67) (0.413) (1.8×10^{-5})
Order of solubility ⇒ Ar < CO_2 < CH_4 < HCHO
Final answer: Option (3)

57)

Explain the Question : We are asked to check the truth of assertion and reason regarding molarity vs. molality.
Concept : This question is based on concentration Units.
Solution :
Assertion : One molar aqueous solution has generally higher concentration than one molal aqueous solution → True (because 1 L of solution contains slightly less solvent than 1 kg of solvent).
Reason : One molar solution contains less solvent than one molal aqueous solution → True, and this is exactly why molarity is higher.
Thus, both are true and the reason correctly explains the assertion.
Final answer : 1

58)

Explain Question: To Calculate the molar mass of non-volatile solute B.

Concept: Relative lowering in vapour pressure.

Solution:
V.P. of Pure solvent A $(P_A^\circ) = 10$ torr
Wt of solute $(w_B) = 1$ g
Wt of solvent $(w_A) = 20$ g
V.P. of solution $(P_s) = 9$ torr
M. wt of solvent (M_A) = 200
M. wt of solute (M_B) = ?

$$\Rightarrow \frac{P^\circ - P_s}{P_s} = \frac{w_B}{M_B} \cdot \frac{M_A}{w_A}$$

$$\Rightarrow \frac{10-9}{9} = \frac{1}{M_B} \cdot \frac{200}{20}$$

$$M_B = 10 \times 9 = 90$$

Final answer: Option (3)

59)

Question Explanation:
Characteristics of non-ideal solutions and Raoult's law.

Concept:
Negative Deviation from Raoult's Law.

Solution:
A solution becomes hot when the mixing process is exothermic ($\Delta H_{mix} < 0$). This happens when the new interactions between different molecules (A-B) are stronger than the original ones (A-A and B-B), leading to a negative deviation. However, stronger interactions pull molecules closer together, causing the volume to decrease ($\Delta V_{mix} < 0$), not increase.
Calculation For Negative Deviation: $\Delta H_{mix} < 0$ (Exothermic/Hot)
$\Delta V_{mix} < 0$
• Volume change: (Decrease in volume)
• The Reason states volume increases, which is false.

Final Answer: Option (3)

60)

Question Explanation: The question tests whether adding a non-volatile solute (NaCl) causes freezing point depression and whether this effect is explained by lowering of vapour pressure.

Concept: Depression in freezing point (a colligative property).

Solution:
Assertion: When NaCl is added to water, freezing point decreases due to presence of solute

particles → **True**.
Reason: Addition of NaCl lowers vapour pressure of water; freezing occurs when vapour pressures of solid and liquid are equal, so lower vapour pressure means freezing at lower temperature → **True**.
Lowering of vapour pressure is the direct cause of freezing point depression → **Correct explanation**.
Both Assertion & Reason are True and the Reason correctly explains the Assertion.

Final Answer: Option (1)

61)

Question Explanation: We are comparing the boiling point elevation of 0.1 m glucose and 0.1 m urea solutions. Both are non-electrolytes and do not dissociate in water.
Concept: Boiling point elevation: $\Delta T_b = i \times K_b \times m$

Solution:
For glucose: i = 1
For urea: i = 1
K_b of water is a constant for the **solvent**, so it is same for both solutions.
Since both have the same i and same m, their ΔT_b must be equal.
Assertion: 0.1 m glucose has higher elevation than 0.1 m urea → False
Reason: K_b is different for both solutions → False (K_b is constant for water)

Both the assertion and reason are incorrect.

Final Answer: Option 4

62)

Explanation the Question:
This question presents an Assertion and a Reason and asks for the correct relationship between the two statements.
Concept:
This question is based on Azeotropes
Solution:
Analyzing the Assertion: The assertion states that azeotropic mixtures are formed by both ideal and non-ideal solutions. This is incorrect. Azeotropes are exclusively formed by non-ideal solutions.
Analyzing the Reason: The reason states that the boiling point of an azeotropic mixture is either higher or lower than both of its components. This is correct, as azeotropes are classified as either minimum boiling or maximum boiling based on this property.
Final Answer: (3)

63) **Question Explanation:** This question tests understanding of factors affecting the rate of evaporation.
Concept : This question is based on Evaporation.
Solution :
Statement I is correct because a larger surface area provides more molecules at the surface

with enough energy to escape into the vapor phase.
Statement II is incorrect. Stronger intermolecular forces hinder the escape of molecules, leading to a slower rate of evaporation.
Statement I is correct but Statement II is incorrect.

Final Answer : option (3)

64) **Explain the Question :** We asked which 1% (w/w) aqueous solution has the highest boiling point
Concept : This question is based on colligative properties-boiling point elevation.
Solution :
For 1% (w/w) solutions (≈ 1g solute in 100 g solution) :

Effective molality $\propto \frac{i}{M_w}$

$CaCO_3$: $M_w \approx 100$, $i = 2 \rightarrow \frac{i}{M_w} = \frac{2}{180} = 0.02$

Urea (CH_4N_2O) : $M_w \approx 60$, $i = 1 \rightarrow \frac{1}{60} \approx 0.0167$

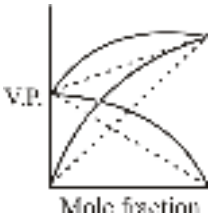
NaCl : $M_w \approx 58.5$, $i = 2 \rightarrow \frac{2}{58.5} \approx 0.034$

$Ca_3(PO_4)_2$: $M_w \approx 310$, $i = 5 \rightarrow \frac{5}{310} \approx 0.0161$

Thus NaCl has the maximum effective particle concentration → greater colligative effect → highest boiling point elevation.
Final Answer: (2)

65)

C_2H_5OH and H_2O solution is non-ideal solution with positive deviation.
$P_S > P_A^\circ X_A + P_B^\circ X_B$
$P_A > P_A^\circ X_A$
$P_B > P_B^\circ X_B$



66)

Question Explanation:
We need to find Henry's law constant (K_h) from the graph of partial pressure (P) versus mole fraction (x) of HCl dissolved in cyclohexane.

Concept:
Henry's law: $P = K_h \times x$

Hence, $K_h = \left(\frac{P}{X}\right)$, which represents the slope of the straight line in the graph.

Solution:
From the graph:
At x = 0.02, P = 1000 torr

So,
$K_h = \left(\frac{P}{X}\right)$ = 1000 / 0.02 = 5×10^4 torr.

Final Answer:
Option 1: 5×10^4 torr

67) **Question Explanation**
This question tests your understanding of how the vapour pressure vs temperature graph changes for a solution vs pure solvent, and how it relates to boiling point elevation.

Concept
A solution has lower vapour pressure than the pure solvent at a given temperature, hence requires higher temperature to boil ⇒ elevation in boiling point.

Solution
From the graph:

A. PQ represents the vapour pressure curve of the solvent.
B. XY lies below it ⇒ vapour pressure of solution.
C. At boiling point, vapour pressure equals atmospheric pressure.
D. Since the solution has lower vapour pressure, it boils at a higher temperature (ΔT= elevation in boiling point).

Hence,

A. PQ : Solvent
B. XY: Solution
C. ΔT : Elevation in boiling point

Final Answer
Option 4

68)

Explain Question:
The question asks for the difference in boiling-point elevation between two sucrose solution of molality 1 m and 2 m, using $K_b = 0.52\ K \cdot kg \cdot mol^{-1}$.

Concept:
Boiling-point elevation: $\Delta T_b = K_b \times m$.

Solution:
For 1 m solution: $\Delta T_1 = 0.52 \times 1 = 0.52$ K

For 2 m solution: $\Delta T_2 = 0.52 \times 2 = 1.04$ K

Difference: $T_2 - T_1 = 1.04 - 0.52 = 0.52$ K

Final answer: Option (3)

69)

Explanation : Determine if the assertion and reason about isotonic solutions and osmosis are true and if the reason explains the assertion.
Concept : Isotonic solutions & osmotic pressure
Solution : Assertion is true: Isotonic solutions have no net osmosis since both sides have equal osmotic pressure.
Reason is also true: Osmotic pressure is the same at constant temperature in isotonic solutions.
The reason directly explains the assertion.
Final Answer : Assertion and reason are true and reason is the correct explanation

70) **Question Explanation:**

Decide correctness of:
Assertion — strong acid + water shows negative deviation from Raoult's law.
Reason — such a solution forms a maximum-boiling azeotrope.

Concept:
Negative deviation (A–B > A–A,B–B) → lower vapour pressure than ideal → higher boiling point at some composition → maximum-boiling azeotrope.

Solution:
Strong acids in water (examples: HCl, HNO_3, H_2SO_4) exhibit strong solute–solvent interactions, giving vapour pressures lower than Raoult's prediction (negative deviation). Solutions that show negative deviation can and do form maximum-boiling azeotropes (e.g., HNO_3–H_2O, HCl–H_2O)..

Final Answer : Option 1.

71)

Question Explanation: The question checks the correctness of the Assertion and Reason related to expressing pollutant concentration in ppm.

Concept: Concentration terms; parts per million (ppm) is used for very dilute solutions.

Solution:

Assertion: Concentration of pollutants in water or air is often expressed in ppm.
→ True, because pollutants are usually present in very small amounts.

Reason: ppm can be expressed as mass/mass, volume/volume, or mass/volume.
→ True, these are valid ways to express ppm.

$$\text{ppm} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$$
$$\text{ppm} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 10^6$$
$$\text{ppm} = \frac{\text{mass of solute}}{\text{volume of solution}} \times 10^6$$

However, the Reason only defines forms of ppm; it does not explain *why* pollutants are expressed in ppm.
Both Assertion & Reason are true, but Reason is not the correct explanation of the Assertion.

Final Answer: Option (2)

72)

Question Explanation: Determine the validity of the assertion and reason for ideal solutions.

Concept: Thermodynamics of Ideal Solutions.

Solution

A. Assertion is true because in an ideal solution, no heat is absorbed or evolved during the mixing process, so $\Delta H_{mix} = 0$.
B. The reason is false because for a solution to be ideal, the A-B intermolecular interactions must be nearly identical to the A-A and B-B interactions.
C. If A-B interactions were less than A-A and B-B, the solution would show a positive deviation from Raoult's law, and ΔH_{mix} would be greater than zero.

Final Answer: Option(3)

73)

Question Explanation: Check validity of Assertion and Reason separately.

Concept: Henry’s Law, Raoult’s Law

Solution:
Henry’s Law: P ∝ mole fraction (valid for dilute solutions)
Raoult’s Law: P = X × P° (ideal solutions)
Raoult’s Law is a **special case** of Henry’s Law when Henry’s constant = vapour pressure

Assertion: Correct
Reason: Correct (definition of Raoult’s law)
Both assertion and reason are correct

Final Answer: Option (4)

74) **A. Question Explanation:**
The question asks for the van't Hoff factor (i) for the association of 3A into A_3, assuming 100% association. The van't Hoff factor represents the number of particles present in solution after the solute dissolves, compared to the number of particles initially dissolved.

B. Given Data:

A. Reaction: $3A \rightarrow A_3$
B. Association: 100%

C. Concept:

A. van't Hoff factor (i) : $\frac{\text{(moles of particles after dissociation or association)}}{\text{(moles of particles before dissociation or association)}}$

Solution:
1. Initial state: Before the reaction, we have 3 moles of A.
2. Final state (100% association): Since the association is 100%, all 3 moles of A combine to form 1 mole of A_3.
3. Calculate the van't Hoff factor:

$$i = \frac{\text{(moles of particles after association)}}{\text{(moles of particles before association)}}$$

$$i = \frac{1}{3}$$

E. Final Answer:
The van't Hoff factor for $3A \rightarrow A_3$ with 100% association is 1/3.

75) **A. Question Explanation:**
The question asks us to calculate the mole fraction of ethanol in rectified spirit, given that it contains 92% ethanol by weight. This means we need to determine the relative number of moles of ethanol compared to the total number of moles in the mixture (ethanol and water).
B. Given Data:

A. Rectified spirit contains 92% ethanol by weight.

C. Concept:

A. **Mole Fraction:**

Solution:
1. Assume a 100g sample: This makes the percentage directly equivalent to grams. So, we have 92g of ethanol and 8g of water.
2. Calculate moles of ethanol:

A. Molar mass of ethanol (C_2H_5OH) = (2 × 12.01) + (6 × 1.01) + 16.00 ≈ 46.08 g/mol
B. Moles of ethanol = (92 g) / (46.08 g/mol) ≈ 2 moles

3. Calculate moles of water:

A. Molar mass of water (H_2O) = (2 × 1.01) + 16.00 = 18.02 g/mol (approximately)
B. Moles of water = (8g) / (18.02 g/mol) ≈ 0.44 moles

4. Calculate total moles:

A. Total moles = Moles of ethanol + Moles of water
B. Total moles ≈ 2 + 0.44 ≈ 2.44 moles

5. Calculate mole fraction of ethanol:

A. Mole fraction of ethanol = (Moles of ethanol) / (Total moles)
B. Mole fraction of ethanol ≈ 2 / 2.44 ≈ 0.818

E. Final Answer:
The mole fraction of ethanol is approximately 0.818.

76)

Explain Question – Calculate the normality by using molarity.

Concept – This question based on Normality = Molarity × basicity.

Solution –
Given Molarity = 2 M, basicity = 1.
Normality = 2 M × 1 = 2 N. The normality of 2 M H_3PO_2 solution will be 2 N

Final Answer – Option (1)

77) **Question Explanation:**
To determine volume strength of H_2O_2 solution.
Formula : $M = \frac{x}{11.2}$
Calculation:
$\frac{w}{m.w \times v_{(\ell)}} = \frac{x}{11.2}$; i = x = ?
$M = \frac{6.8}{34} \times \frac{1}{1} = 0.2$
Vol. strength = M × 11.2
= 0.2 × 11.2 = 2.24 V
Answer is option (3).

78) **Question Explanation:** This question applies Raoult's Law for non-volatile solutes, which states that the vapor pressure of a solution is directly proportional to the mole fraction of the solvent.
Concept : This question is based on Raoult's Law
Solution :
According to Raoult's Law: $P_{solution} = X_{solvent} \times P^0_{solvent}$
Here, mole fraction of solvent (water) $X_{water} = 1 - 0.10 = 0.90$.
$P^0_{water} = 100$ mm Hg
$P_{solution} = 0.90 \times 100 = 90$ mm Hg.

Final Answer : option (3)

79) **Explain Question:** To calculate the boiling point of 1.0 molal KCI solution.
Concept: Elevation in Boiling point (for i > l)
Solution:
$K_b = 0.52$ K kg mol^{-1}
Molality = 1.0 m
For KCI solution (i = 2)
$\Delta T_b = i \times K_b \times m$
$\Delta T_b = 2 \times 0.52 \times 1.0$

= 1.04

$\Delta T_b = T_b - T^o_b$

$T_b = T^o_b + \Delta T_b$

= 100 + 1.04 = 101.04 ºC

Final answer: (2)

80) **Explain Question :** Find moles of NaCl from relative lowering of vapour pressure.

Concept : Relative lowering

Solution : $\Delta P/P° = i(n_{solute} / n_{solute + solvent})$

For NaCl, i = 2.

Moles of H_2O = 180/18 = 10 mol.

0.18 = 2n / (2n + 10)

0.36n + 18 = 2n

1.64n = 18

n = 1.1 mol

Final Answer: Option (2)

81) **Explain the question :** We need to find equilibrium constant (K_C) for given reaction.

Concept : This question is based on equilibrium constant (K_C)

Solution :

	A	+	B	⇌	C	+	D
t = 0	1		1		–		–
t_{eq}	1 – x		1 – x		x		x

At equilibrium : [C] = 2[A]

x = 2 (1 – x) = 2 – 2x

3x = 2

$x = \frac{2}{3}$

$1 - x = 1 - \frac{2}{3} = \frac{1}{3}$

$$K_C = \frac{[C][D]}{[A][B]} = \frac{\frac{2}{3} \times \frac{2}{3}}{\frac{1}{3} \times \frac{1}{3}} = 4$$

$K_C = 4$

Final answer : 1

82)

Question Explanation : We need to compare K_p and K_c. Relation is

$K_p = K_c (RT)^{\Delta n}$

If $\Delta n < 0 \rightarrow K_p < K_c$.

Concept:- This question is based on Δn = moles of gaseous products - moles of gaseous reactants.

Solution :-

1. $N_2O_3(g) \rightleftharpoons NO_2 + NO$

$\Delta n = (1 + 1) - 1 = +1 \rightarrow K_p > K_c$

2. $2H_2 + O_2 \rightleftharpoons 2H_2O$

$\Delta n = 2 - (2 + 1) = -1 \rightarrow K_p < K_c$

3. $H_2 + I_2 \rightleftharpoons 2HI$

$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$

4. $2SO_3 \rightleftharpoons 2SO_2 + O_2$

$\Delta n = (2 + 1) - 2 = +1 \rightarrow K_p > K_c$

Final Answer : Option (2)

$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

83)

Explaining the Question :- We are comparing the stability of nitrogen oxides using their equilibrium constants of formation.

Concept:- This question is based on Larger equilibrium constant → product is more stable (formation favored).

Solution :- For NO: $K = 5 \times 10^{-31}$

• For NO_2 ; $K = 8 \times 10^{-31}$

• For N_2O_5 ; $K = 1 \times 10^{-34}$

Order of stability (higher K → more stable)

NO_2 (8×10^{-31}) > NO (5×10^{-31}) >> N_2O_5 (10^{-34})

Final Answer :- 4

84) **Explain the Question :** We need to find effect of concentration on equilibrium constant.

Concept : This question is based on equilibrium constant depends only on temperature.

Solution : At constant temperature, K_{eq} will be constant and does not depend on concentration.

Final Answer : 3

85)

Explain Question:

To find out K_b.

Concept:

Equilibrium constant.

Solution:

$$K_{eq} = \frac{K_f}{K_b} \Rightarrow 20 = \frac{10}{K_b}$$

$$K_b = \frac{10}{20}$$

$$= \frac{1}{2} = 0.5$$

Final answer: Option (1)

86) $NaN_3 \rightarrow Na^+ + N_3^-$ (i = 2)

Let a gm ice form

wt of solvent in liquid state (1000 –a)g

$$\Delta T_f = iK_f \frac{n_B \times 1000}{w_A(g)}$$

$$10 = \frac{2 \times 1.8 \times 2 \times 1000}{(1000 - -a)}$$

1000 –a = 720

a = 280 g

87) **Explain Question:** Assertion (A): the boiling point of 0.1 M urea solution is less than of 0.1 M KCI solution.

Reason (R): Elevation of boiling point is directly proportional to the number of moles of non-volatile solute particles present in the solution.

Concept: KCI dissociates (i = 2), giving more particles than urea (i = 1) . More particles → greater elevation → higher boiling point.

Solution: Assertion (A) is true. Reason (R) is also true and correctly explains (A).

Final Answers: Option (4)

88) NCERT Pg. # 39

89) **Explain question:**

To find the concentrated or dilute solutions among A and B.

Concept:

Osmosis and Osmatic Pressure.

Solution:

Osmosis It is defined as spontaneous net flow of solvent molecules through a semi permeable membrane form a region of lower solute concentration to Higher solute concentration.

So, if solute flows from A to B then A is more dilute than B.

Final Answer: Option (1)

90)

Explain the Question : Compare the osmotic pressure of a 2M acetic acid solution (100% dimerized) and a 0.5 M KCl and a 0.5 M KCl solution (100% ionized).

Concept : This question is based on osmotic pressure π = iCRT,

Solution :

1. Acetic Acid : $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$.

• Since 100% dimerization occurs, two molecules form one. Thus, i = 0.5.

• Effective concentration = i × C = 0.5 × 2 m = 1 m

2. KCl → $K^+ + Cl^-$

• 100% ionization means each molecules splits into two ions. Thus, i = 2

• Effective concentration = i × C = 2 × 0.5 M = 1 M.

Both solutions have the same effective concentration (1M) and therefore, the same osmotic pressure.

Final Answer : (1)

BIOLOGY

91)

NCERT XII, Page # 128

92) **Question Explanation:** Meteorite analysis and chemical evolution.
Concept: Extraterrestrial origin of life.
Solution: The **theory of chemical evolution** suggests that organic molecules, the building blocks of life, could have formed under primitive Earth conditions. The discovery of similar organic compounds on meteorites, like the **Murchison meteorite**, provides evidence that the processes leading to the formation of these molecules are not unique to Earth but can occur elsewhere in the universe. This supports the idea that life's building blocks could have also come from space.
Final Answer: Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

93)

NCERT-XII, Pg. # 111

94)

NCERT (XII) Pg # 128/137(H) Para : 7.1

95)

A. Query successful

Question Explanation: Matching major events to their timeline.**Concept:** Timeline of Evolution
Solution: Here's the correct timeline matching:

A. (A) The **Origin of the Universe** (Big Bang) occurred approximately **20 billion years ago** (iv).
B. (B) The **Origin of Earth** occurred about **4.5 billion years ago** (ii).
C. (C) The first **cellular forms of life** appeared around **2 billion years ago** (v).
D. (D) The first **non-cellular forms of life** (e.g., RNA, proteins) appeared around **3 billion years ago** (iii).
E. (E) **Life appeared** about **4 billion years ago** (i), as non-cellular forms.

Final Answer: A - iv, B - ii, C - v, D - iii, E - i

96) NCERT-Pg. # 111

97) NCERT Page No. 127

98) NCERT Pg. # 111 (E) / 123 (H)

99) NCERT-XII Page No. 111

100)

Module

101) NCERT XII (E) Pg # 127

102) NCERT XII, Pg.# 127

103) NCERT (XIIth) Pg. # 128(E); Pg. # 138 (H)

104) NCERT XII pg.# 137, para 4

105) NCERT XII, Page # 127

106)

NCERT Pg # 127

107) NCERT-XII Pg# 127

108) NCERT (XIIth) Pg. # 127

109)

Theory of special creation

110)

Explain Question: Match each disease with its correct causative agent or characteristic symptom.

Concept: This Question is based on Protozoan diseases, Viral Diseases, Bacterial Diseases

Explanation :

Common cold (A) → Rhino virus (iv)

Typhoid (B) → Salmonella typhi (v)

Pneumonia (C) → Haemophilus influenzae (ii)

Malaria (D) → Haemozoin (vi)

Ringworm (E) → Dry & Scaly lesions (i)

Amoebiasis (F) → Stool with mucous & blood clots (iii)

This matches with : A–iv, B–v, C–ii, D–vi, E–i, F–iii **Final answer:** 1

111)

Question Explanation: Choose which row correctly identifies A, B, C, D in the life cycle in the malaria.

Concept: Malaria

Solution: In the malaria (Plasmodium) life cycle sporozoites enter the host and go to the liver; later stages (gametocytes) are formed in the host blood and sporozoites are produced in the mosquito salivary gland. So ,

A — Sporozoites

B — Liver

C — Salivary gland

D — Gametocyte

Final answer: Option 3

112)

Explain Question : To identify the total number of correct statements about the given diagram.

Concept : This question is based on Acquired immunity -Antibody.

Solution :
This is a diagram of an antibody molecule
(a) Gives cell mediated immunity **Incorrect**
(b) T-cells themselves do not secrete but help
B-cell to produce them. **correct**
(c) Disulphide bond present **correct**
(d) H_2L_2 molecule **correct**
(e) Glyco protein molecule **correct**
So,4 statements are correct

Final Answer : option (4)

113)

Question Explanation –
The question asks to correctly fill the missing disease names, causative organisms, and symptoms in the given table by matching them with known human diseases and their characteristics.
Concept – This question is based on Human diseases and pathogens
Solution –

A. Ascariasis is caused by Ascaris and is associated with symptoms like internal bleeding, muscular pain, fever, anemia, and blockage of intestinal passage.

B. Trichophyton causes ringworm, which shows dry, scaly lesions on various parts of the body.

C. Common cold is caused by rhinovirus and shows symptoms like nasal congestion, discharge, sore throat, cough, and headache.

D. Filariasis is caused by Wuchereria, leading to inflammation of lower limbs.

Only Option 2 correctly matches the missing entries (B – Ringworm and D – Wuchereria) without any incorrect pairing.
NCERT (XIIth) Pg. # 149
Final Answer – Option 2

114) NCERT-XII, Pg.#147, 148

115)

Question explanation: Assertion and reason based

Concept: Immunity

Solution: Assertion (A) :– Everyday we are exposed to large number of infectious agents. However only few of these exposures result in disease-correct

Reason (R) :– Our body is able to defend itself from most of these foreign agents. This overall ability of host to fight the disease causing organism conferred by immune system is called immunity-correctBoth Assertion & Reason are True & the Reason is a correct explanation of the Assertion
Final answer: 1

116) NCERT XII Pg # 153

117) NCERT XII Pg # 159

118) NCERT Pg # 157 [E]
NCERT Pg # 172 [H]

119) NCERT XII Page No. # (E)149, (H) 160

120) NCERT Pg. # 154

121) NCERT - XII Pg. # 162 - 154

122) The figure shows lymphoid organs labeled A, B, and C. The task is to identify which label corresponds to a primary lymphoid organ.
Concept: The question is based on immune system.
Solution:

A. Primary lymphoid organs: Bone marrow and thymus (sites of lymphocyte production and maturation).
B. Secondary lymphoid organs: Lymph nodes, spleen, tonsils, Peyer's patches, etc.
C. In the given labels:
    A. A → Thymus → Primary lymphoid organ.
    B. B → Lymphatic vessels → Not a lymphoid organ, just transport channels.
    C. C → Lymph nodes → Secondary lymphoid organ.

Thus, only A (Thymus) is the primary lymphoid organ.

Final answer: **Option 1**

123)

Explain Question : TO IDENTIFY THE GIVEN DIAGRAM
Concept : This question is based on DRUG ABUSE
Explanation : THE GIVEN DIAGRAM SHOWS THE PART OF PLANT, DHATURA
Final answer : 2

124) NCERT Pg No. # 151 (Fig. 8.4)

125) NCERT-XII, Pg.#149, Para.-3

126)

Explaining the Question :- Which statement is correct about the disease shown in the diagram ?
Concept:- This question is based on Fungal skin infection (Ringworm/Dermatophytosis)

Solution :- Plasmodium → causes malaria, not skin lesions
Wuchereria bancrofti → causes filariasis
Allergic reaction → does not produce typical ring-shaped scaly lesions.
Heat and moisture help fungal growth
Final Answer :- 3

127)

Explaining the Question :- Which statement about drug/alcohol abuse is false ?
Concept:- This question is based on Effects of drug and alcohol abuse

Solution :-
• Statement 4 is false
Withdrawal symptoms can be severe and even life-threatening, especially with substances like alcohol and opioids (e.g., delirium tremens, seizures)
All other statements correctly describe the harmful effects of drug/alcohol abuse.
Final Answer :- 4

128) NCERT-12th Pg No. # 132, 133

129) NCERT (XII) Pg. # 150, Para–4

130)

Question Explanation: The question asks how many of the given disease–causation statements are correct.

Concept: Infectious diseases

Solution:

A. Malignant malaria is caused by Plasmodium vivax → Incorrect
(correct: Plasmodium falciparum)

B. Typhoid caused by Haemophilus influenzae → Incorrect
(correct: Salmonella typhi)

C. Common cold is caused by Rhinovirus → Correct

D. Amoebiasis is caused by bacteria → Incorrect
(correct: Entamoeba histolytica, a protozoan)

Only one statement is correct.

Final answer: Option 2

131) NCERT (XII) Pg. # 145.146

132)

NCERT Pg. No. 138

133)

Explain Question:
This question asks to count how many of the listed diseases are infectious diseases.

Concept
Infectious diseases.

Solution:

A. Infectious diseases are caused by pathogens/Infectious agents like viruses, bacteria, etc.
B. (i) Small pox → infectious (viral)
C. (ii) Allergy → non-infectious
D. (iii) Influenza → infectious (viral)
E. (iv) AIDS → infectious (viral)
F. (v) Cancer → non-infectious

So, Infectious diseases are Small pox, Influenza, and AIDS = 3 diseases.

Final Answer: Option (3)

134)

Explain Question: The question asks which factors affect our health.

Concept: Health and disease

Solution: The correct answer is Option 4.

A. Health is influenced by **genetic disorders**, **infections**, and **lifestyle** factors such as diet, exercise, and habits.

Final Answer: Option 4

135)

Question explanation: Smoking affects multiple organs and is associated with cancers, respiratory, cardiovascular, and digestive disorders.
Concept: This question is based on smoking

Solution: The correct answer is option 4 → a, b, c, d, e, f, g

A. **(a) Lung cancer** – Strongly linked to smoking.

B. **(b) Bronchitis** – Smoking irritates airways causing chronic bronchitis.

C. **(c) Emphysema** – Smoking damages alveoli leading to emphysema.

D. **(d) Coronary heart disease** – Smoking increases risk of heart diseases.

E. **(e) Gastric ulcer** – Smoking increases acid secretion and delays healing.

F. **(f) Urinary bladder cancer** – Carcinogens in smoke are excreted via urine.

G. **(g) Throat cancer** – Smoking damages oral and throat tissues.

Final Answer: Option 4

136)

NCERT (XIIth) Pg. # 133

137)

NCERT (XIIth) Pg. # 137,138

138) NCERT Pg # 151-152

139) NCERT Pg # 157

140) **Question Explanation:** Matching types of immunity.
Concept: Types of Immunity.
Solution: Active natural immunity results from **Infection** (i→d). **Active artificial** immunity is acquired via **Vaccination** (ii→c). **Passive natural** immunity is gained from the mother, e.g., **Colostrum** (iii→b). **Passive artificial** immunity involves receiving preformed antibodies, e.g., **Antiserum** (iv→a).
Final Answer: d c b a

141) **Question Explanation –** The question is asking to correctly match different classes of antibodies with their characteristic features.

Concept – Classes of antibodies and their functions

Solution –

IgG → Most abundant antibody → (iii)
IgM → First antibody to appear in primary immune response → (ii)
IgA → Found in saliva and other secretions → (iv)
IgE → Involved in allergic & parasitic infections → (i)

Correct matching: A-(iii); B-(ii); C-(iv); D-(i)

Final Answer – Option [1]

142)

NCERT XII Pg. # 141 (E), 155 (H)

143) NCERT (XIIth) Pg. # 153,154

144) **Question Explanation:** This question asks to identify the disease that is not transmitted by a vector.
Concept: Vector-borne diseases
Solution: The correct answer is (Option 3.)
Pneumonia – Usually caused by bacteria (like Streptococcus pneumoniae) or viruses; it spreads via air droplets from coughs or sneezes, not by a vector.
Dengue, Chikungunya, and Malaria are vector-borne (transmitted by mosquitoes)
Final Answer: Pneumonia

145)

Question Explanation: Identify the incorrect match between condition/type and its example.

Concept: Human Health and Disease – Immunity and Disorders

Solution: The correct answer is option: **2**

A. **Allergy → Asthma** - Correct

B. **Autoimmune disorder → Down syndrome** - Incorrect

    A. Down syndrome is a **genetic disorder**, not an autoimmune disorder.

C. **Active immunity → Vaccination** - Correct

D. **Passive immunity → IgA in colostrum** - Correct

Final answer: option 2

146)

Explain Question : to identify the lightest antibody

Concept : This question is based on antibody.

Solution : IgG is the Lightest antibody it has lowest molecular weight.

Final Answer : option (1)

147) NCERT (XII) Pg. # 150,151, Para 8.2.1

148) NCERT–XI, Pg. # 148, Para-8.1

149) NCERT Pg. # 151

150) NCERT (XII) Pg # 154

151) NCERT-XII, Pg#146, last line

152) NCERT-XII, Pg#152, 2nd para

153) NCERT Page # 148 (E)

154)

Module

155) NCERT-XII, Pg # 146

156) NCERT (XIIth) Pg. # 145

157) NCERT (XIIth) Pg. # 146

158) Malaria is a protozoan disease, caused by *Plasmodium* species. In case of malaria, there is release of a toxic substance(called haemozoin) from the ruptured RBCs which is responsible for the chill and high fever.

159) **Explanation Question :**
The question is asking whether Pneumonia is caused by Haemophilus influenzae and whether it affects the alveoli of lungs causing respiratory issues.
Concept: This question is based on human health and disease – Pneumonia.
Solution:
Pneumonia is mainly caused by Streptococcus pneumoniae and Haemophilus influenzae also contributes, so Statement-I is correct.
Pneumonia infects the alveoli of lungs causing accumulation of fluid and breathing difficulty, so Statement-II is also correct.
Final answer: 1

160) NCERT XII Pg.# 130 / 143 (H), Para : 7.1

161) NCERT XII Pg.# 138 / 151 (H), Para : 7.2.7

162) NCERT XII Pg.# 134 / 147 (H), Para : 7.2

163) NCERT XII Pg.# 135 / 147 (H), Para : 7.2.2

164) NCERT XII Pg.# 135 / 148 (H), Para : 7.2.2

165) NCERT XII Pg.# 137 / 150 (H), Para : 7.2.5

166) NCERT XII Pg.# 141 / 155 (H), Para : 7.4

167) NCERT XII Pg.# 143 / 157 (H), Para : 7.5

168) NCERT XII Pg.# 131 / 143 (H), Para : 7.1

169) NCERT XII Pg.# 131 / 143 (H), Para : 7.1

170) NCERT XII Pg.# 133 / 145 (H), Para : 7.1

171) NCERT XII Pg.# 136 / 143 (H), Para : 7.1

172) NCERT XII Pg.# 140 / 153 (H), Para : 7.3

173) NCERT XII Pg.# 141 / 154 (H), Para : 7.4

174)

NCERT XII Pg.# 142/156 (H) ; Para : 7.5

175) NCERT XII Pg.# 138

176) NCERT XII Pg.# 141

177)

NCERT XII Pg.# 142

178) NCERT XII Pg.# 131

179) NCERT XII Pg.# 137

180) NCERT XII Pg.#131-143